MODELING THE AIR QUALITY IMPACT OF STUBBLE BURNING ON THE NATIONAL CAPITAL REGION (NCR) USING MULTI-SENSOR REMOTE SENSING AND MACHINE LEARNING

Project Report

Study Period: 2018 – 2023

Burning Season: September – December (Annual)

Data Sources:

MODIS Terra/Aqua | Sentinel-5P TROPOMI | ERA5 Reanalysis | NOAA HYSPLIT

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1. INTRODUCTION

Stubble burning, the practice of setting fire to crop residues after harvest, has emerged as one of the most critical drivers of seasonal air quality deterioration in northern India. Every post-kharif season between October and November, farmers in Punjab and Haryana burn millions of tonnes of paddy straw to clear fields rapidly for the ensuing rabi crop cycle. The resulting smoke plumes, laden with particulate matter, carbon monoxide (CO), nitrogen dioxide (NO_2), and carbonaceous aerosols, are transported hundreds of kilometres to the south-east by prevailing north-westerly winds, routinely enveloping the National Capital Region (NCR) — comprising Delhi and its satellite cities — in a thick, toxic haze. The consequences for public health, aviation, and the local economy are severe and well-documented, yet a rigorous, data-driven characterization of the fire–pollution linkage at daily temporal resolution has remained elusive.

1a. Literature Survey

The scientific literature on stubble burning and its atmospheric impacts spans several disciplines including satellite remote sensing, atmospheric chemistry, epidemiology, and machine learning. Collectively, these studies establish the broad physical basis for the relationship investigated in the present work.

Remote Sensing of Fire Activity:

The Moderate Resolution Imaging Spectroradiometer (MODIS) aboard NASA's Terra and Aqua satellites has been the cornerstone of fire monitoring over the Indo-Gangetic Plain (IGP). The MOD14A1 active fire product detects thermal anomalies at 1 km spatial resolution by exploiting mid-infrared and thermal-infrared channels, providing daily fire counts and Fire Radiative Power (FRP) estimates. The complementary MCD64A1 burned area product, derived from surface reflectance change detection at 500 m resolution, provides monthly estimates of the spatial extent of burning. Kumar et al. (2020) used MODIS data to demonstrate a consistent north-westerly transport pathway of biomass burning emissions from Punjab and Haryana to NCR during the post-kharif season. Vadrevu et al. (2018) established the inter-annual variability in stubble burning intensity and its dependence on paddy harvest timing, policy interventions, and meteorological conditions.

Sentinel-5P TROPOMI and Trace Gas Retrievals:

The TROPOspheric Monitoring Instrument (TROPOMI) aboard ESA's Sentinel-5 Precursor satellite, operational since November 2017, provides daily global maps of trace gas column concentrations at unprecedented spatial resolution (~3.5 × 5.5 km per pixel at nadir). Sentinel-5P data products including tropospheric NO_2, CO total column, and UV absorbing aerosol index (AAI) have been widely used to characterize biomass burning outflows. A study published in MDPI's Remote Sensing (2022) demonstrated spatiotemporal correlations between stubble burning activity and elevated TROPOMI-derived pollutant columns over NCR during the October–November burning window. Javadi et al. used Sentinel-5P NO_2 data combined with machine learning to develop a spatiotemporal prediction framework for Delhi's atmospheric pollution. Singh and Pant (2024) similarly used TROPOMI to trace haze episodes to source regions in Punjab and Haryana via satellite-based aerosol back-trajectory analysis.

Atmospheric Transport Modelling — HYSPLIT:

The NOAA Hybrid Single-Particle Lagrangian Integrated Trajectory (HYSPLIT) model is the international standard tool for atmospheric back-trajectory analysis and source attribution. It computes the three-dimensional path of air parcels backward in time from a receptor point using meteorological fields such as GDAS (Global Data Assimilation System) or ERA5 reanalysis. Numerous studies have employed HYSPLIT to establish the causal link between Punjab–Haryana stubble burning and pollution episodes in Delhi. Sarkar et al. (2018) computed 72-hour HYSPLIT back-trajectories arriving at Delhi on high-pollution days and found that 60–75% of trajectories during October–November originated from the north-west, passing directly through the fire-active regions of Punjab and Haryana. This finding has been repeatedly corroborated and forms the physical backbone of the transport hypothesis tested in the present study.

Machine Learning for Pollution Prediction:

Recent years have seen growing application of ensemble machine learning methods — particularly Random Forest, Gradient Boosting, and XGBoost — for atmospheric pollution prediction. These non-linear models are well-suited to capture complex interactions between fire activity, meteorology, and pollution concentrations. Multiple studies have reported that incorporating meteorological variables (wind speed, boundary layer height, temperature) substantially improves predictive performance relative to fire-only models. The planetary boundary layer height (BLH), in particular, exerts a dominant influence on surface pollutant concentrations by controlling the atmospheric volume available for mixing. Low BLH values during October–November inversions have been shown to amplify pollution severalfold even with moderate emission strengths.

Policy Context:

The Indian government has progressively tightened regulations on open burning under the National Clean Air Programme (NCAP) and the Commissions for Air Quality Management (CAQM) guidelines. Despite financial incentives for in-situ residue management through the Happy Seeder and similar machinery, compliance remains incomplete. TERI (2018) estimated health costs of stubble burning to be ₹500–800 crore per burning season. The World Bank (2020) estimated that air pollution in South Asia costs approximately 7.4% of regional GDP annually. Against this backdrop, scientifically rigorous quantification of the fire–pollution relationship is essential for evidence-based policymaking.

1b. Objectives of the Study

The specific objectives of this study are:

1. To characterize the inter-annual and intra-seasonal variability of fire activity (fire count, FRP, burned area) in Punjab and Haryana during the post-kharif burning season from 2018 to 2023 using MODIS satellite products.
2. To analyse the temporal evolution of air pollution metrics (CO, NO_2, Aerosol Index) over NCR during the same period using Sentinel-5P TROPOMI data and compute seasonal anomalies relative to pre-season baselines.
3. To quantify the statistical relationships between fire activity anomalies in Punjab–Haryana and pollution anomalies in NCR using Pearson correlation analysis, regression modelling, and non-parametric hypothesis testing.

4. To develop and compare machine learning models — Linear Regression, Random Forest, and XGBoost — with and without temporal lag features, to predict daily pollution anomalies from fire-related variables.
5. To investigate the additional explanatory power of ERA5 meteorological variables (boundary layer height, wind speed, temperature, humidity, precipitation) in improving model predictive performance.
6. To implement a probabilistic HYSPLIT back-trajectory simulation framework and validate it against a real NOAA HYSPLIT trajectory, in order to establish the physical transport pathway from Punjab–Haryana to NCR.
7. To perform a conceptual TROPOMI vertical column analysis for source attribution to distinguish fire-transported pollution from local surface emissions over NCR.
8. To formulate a proxy Burn Severity Index (NBR proxy) as a physically interpretable indicator of fire intensity in the absence of direct spectral NBR data.

1c. Study Area

The study encompasses two primary geographic zones, selected based on their ecological and functional roles in the fire–pollution system:

Source Region — Punjab and Haryana:

Punjab and Haryana form the agricultural heartland of India and together account for approximately 8–10 million tonnes of paddy straw generated annually. Punjab alone contributes roughly 60% of the national paddy stubble burning incidents detectable from satellites. Geographically, the study area is bounded by approximately 29.5°N–32.5°N latitude and 73.5°E–77.5°E longitude, encompassing all major paddy-producing districts. The flat alluvial topography of the region, combined with intense mechanized harvesting from October onwards, creates a compressed burning window and high spatial density of fire events.

Receptor Region — National Capital Region (NCR):

The National Capital Region includes Delhi and the surrounding districts of Haryana (Gurgaon, Faridabad, Sonepat, Rohtak, Jhajjar, Palwal, Nuh, Rewari, Panipat, Mahendragarh, Bhiwani, Charkhi Dadri, Karnal, Jind), Uttar Pradesh (Ghaziabad, Gautam Buddha Nagar, Meerut, Baghpat, Hapur, Bulandshahr, Muzaffarnagar, Shamli), and Rajasthan (Alwar, Bharatpur). With a population exceeding 47 million, NCR is the world's largest urban agglomeration and already faces severe ambient air quality challenges from vehicular emissions, industrial activity, and construction dust. The burning season overlay amplifies these pre-existing stressors dramatically.

Transport Corridor:

The two regions are connected by a north-west to south-east transport corridor of approximately 300–500 km. Under the dominant October–November wind regime, north-westerly winds at 3–6 m/s carry smoke aerosols from Punjab and Haryana toward NCR in approximately 2–5 days. Shallow atmospheric boundary layers during this season (often below 800 m) trap pollutants close to the surface, dramatically amplifying ground-level concentrations.

Figure 1. Study Area Map showing Punjab-Haryana Source Region and NCR Receptor Region

2. MATERIALS AND METHODS

2a. Data Sources

This study integrates multiple satellite and reanalysis datasets retrieved via Google Earth Engine (GEE) and the NOAA HYSPLIT web interface. Table 1 summarizes all primary data sources.

Dataset	Satellite/Source	Variable(s)	Spatial Res.	Temporal Res.	Period
MODIS MOD14A1	Terra (NASA)	Fire Count, MaxFRP	1 km	Daily	2018–2023
MODIS MCD64A1	Terra+Aqua (NASA)	Burned Area	500 m	Monthly	2018–2023
Sentinel-5P TROPOMI	ESA Copernicus	CO, NO_2, Aerosol Index	~3.5×5.5 km	Daily	2018–2023
ERA5 Reanalysis	ECMWF	BLH, Wind, Temp, RH, Precip	31 km	Hourly/Daily	2018–2023
FAO GAUL 2015	FAO	Administrative Boundaries	Vector	Static	2015
GDAS Reanalysis	NOAA/NCEP	Meteorology for HYSPLIT	~50 km	6-hourly	2026 (validation)

Table 1. Summary of Data Sources Used in the Study

2b. Data Preprocessing and Dataset Construction

Fire Data Extraction (MOD14A1):

Fire counts were extracted from the MOD14A1 FireMask band by counting pixels with confidence level ≥7 (i.e., fire pixels with high confidence). FRP was extracted from the MaxFRP band and summed over the Punjab–Haryana source region on a daily basis. All fire statistics were computed using GEE's reduceRegion() function with a scale of 1000 m.

Burned Area Extraction (MCD64A1):

Monthly burned area was extracted from the BurnDate band by counting pixels with BurnDate > 0 (indicating confirmed burning events). Pixel count was multiplied by pixel area to obtain total burned area in square metres, subsequently converted to km^2.

Pollution Data Extraction (Sentinel-5P TROPOMI):

Daily mean CO column number density, tropospheric NO_2 column number density, and UV absorbing aerosol index were extracted as spatial means over the NCR target region using GEE's reduceRegion() with a mean reducer at 1000 m scale. Data were collected from three separate TROPOMI Level-2 collections: COPERNICUS/S5P/OFFL/L3_CO, COPERNICUS/S5P/OFFL/L3_NO2, and COPERNICUS/S5P/OFFL/L3_AER_AI.

Temporal Scope:

Data were collected from June through December for each year from 2018 to 2023. The June–July period served as the pre-season baseline, while September–December constituted the burning season analysis window. Daily observations were merged on date key to form a unified dataset. After aggregation and cleaning, the final dataset contained daily records covering six burning seasons.

2c. Anomaly Computation Methodology

A year-specific baseline approach was adopted to compute anomalies. For each year, the mean values of fire count, total FRP, and burned area during June–July (the baseline period, characterized by minimal stubble burning activity) were computed. Similarly, mean values of CO, NO_2, and aerosol index during June–July were computed for each year as pollution baselines.

Anomalies were then defined as the departure of each daily burning-season observation from the corresponding year-specific baseline:

$$\textit{Anomaly} = \textit{Seasonal Value} - \textit{Year-specific Baseline Mean}$$

This approach isolates the signal attributable to burning-season dynamics from the background annual cycle. Missing fire data were filled with zeros (consistent with no-fire conditions), burned area gaps were forward-filled within months (to account for the monthly MODIS product frequency), and pollution data gaps were interpolated linearly.

2d. Proxy Burn Severity Index (NBR Proxy)

The standard Normalized Burn Ratio (NBR) requires near-infrared (NIR) and shortwave-infrared (SWIR) spectral bands, which were not directly available in the dataset. To provide a physically interpretable burn severity indicator, a proxy index was computed from the available MODIS fire variables:

$$\text{NBR Proxy} = \text{Burned Area} / (\text{Fire Count} + \varepsilon)$$

where $\varepsilon = 10^{-6}$ is a small constant to avoid division by zero. This proxy captures the average burned area per fire event, serving as a measure of fire spatial extent relative to fire count. High proxy values indicate fewer but spatially extensive fires (suggesting intense, sweeping combustion), while low values indicate numerous small fires. The proxy was scaled to the range [−1, +1] for physical interpretability, analogous to spectral NBR values where near +1 indicates healthy vegetation and near −1 indicates severe burning.

2e. Lag Feature Engineering

Atmospheric transport from Punjab–Haryana to NCR requires finite time, during which pollutants are dispersed, chemically transformed, and mixed with background air. To account for this temporal delay, lagged versions of fire activity variables were created by shifting daily observations by 0 to 5 days:

- fire_anomaly_lag_N: fire count anomaly shifted by N days (N = 0, 1, 2, 3, 4, 5)
- frp_anomaly_lag_N: FRP anomaly shifted by N days (N = 0, 1, 2, 3, 4, 5)

The lagged dataset was saved separately after dropping rows with missing values introduced by the shifting operation. These lag features were incorporated into machine learning models to test whether temporal information improves prediction of pollution anomalies relative to same-day fire activity alone.

2f. Machine Learning Modelling Framework

Three machine learning algorithms were employed for regression modelling of pollution anomalies from fire activity variables:

Linear Regression:

A baseline model implemented using scikit-learn's LinearRegression class. Features: fire_anomaly, frp_anomaly, burn_anomaly (no-lag) and fire/frp lag features (with-lag). Train/test split: 80%/20%, random_state=42.

Random Forest Regressor:

An ensemble of decision trees implemented using RandomForestRegressor with n_estimators=200-300, max_depth=6-8, random_state=42. RandomForest is particularly effective for non-linear, high-dimensional relationships and provides feature importance scores.

XGBoost Regressor:

Gradient-boosted trees implemented using XGBRegressor with n_estimators=300-500, max_depth=5-6, learning_rate=0.03-0.05, subsample=0.8-0.9, colsample_bytree=0.8-0.9. XGBoost is typically robust against overfitting through regularization.

Model performance was evaluated using R^2 (coefficient of determination) and RMSE (root mean squared error). A TimeSeriesSplit cross-validation approach was also applied in the meteorological integration phase to ensure temporal generalization.

2g. HYSPLIT Back-Trajectory Analysis

The NOAA HYSPLIT (Hybrid Single-Particle Lagrangian Integrated Trajectory) model was used for atmospheric source attribution. Two complementary approaches were implemented:

Probabilistic Simulation Framework:

A Python-based simulation engine was developed to generate 72-hour backward trajectories from NCR/Delhi (receptor: 28.65°N, 77.23°E) at an arrival height of 500 m AGL. Trajectory direction probabilities were drawn from seasonal wind climatology, with a 55% north-westerly probability during October–November based on ERA5/IMD climatological statistics. For each simulated trajectory, a Boolean flag was assigned indicating whether the trajectory intersected the Punjab–Haryana fire bounding box (29.5°–32.5°N, 73.5°–77.5°E). Simulated pollution values (NO_2, aerosol index) were assigned probabilistically as a function of trajectory origin to test the source attribution hypothesis.

Real NOAA HYSPLIT Trajectory (Validation):

A single reference trajectory was generated using the NOAA HYSPLIT web interface with real GDAS (Global Data Assimilation System) reanalysis meteorological data. Receptor: Delhi (28.65°N, 77.23°E), arrival height: 500 m AGL, trajectory duration: 72 hours backward. This trajectory served to validate the physical plausibility of the north-westerly transport pathway assumed in the simulation, independent of the burning season dates.

2h. TROPOMI Vertical Column Analysis

A conceptual vertical profile analysis was performed to develop a framework for distinguishing fire-transported pollution from local surface-emitted pollution in TROPOMI column data. Synthetic NO_2 vertical profiles were constructed for three scenario types: (1) fire-influenced days (enhanced concentration at 1.5–5.6 km altitude, consistent with mid-tropospheric transport of biomass burning smoke), (2) local pollution days (enhanced near-surface concentration below 1 km, consistent with urban traffic and industrial sources), and (3) clean background days (uniformly low concentrations). Total column values were compared across scenarios to characterize the observational signature of each pollution type.

2i. ERA5 Meteorological Integration

ERA5 hourly reanalysis data were aggregated to daily values for the NCR bounding box (28°–29°N, 76.5°–78°E) and appended to the main analysis dataset. Variables included:

- Boundary Layer Height (BLH, m): controls the atmospheric mixing volume
- 10 m Wind Speed (m/s): governs horizontal pollutant transport
- 2 m Air Temperature (°C): influences atmospheric stability and chemistry
- Relative Humidity (%): affects aerosol hygroscopicity
- Daily Precipitation (mm): governs wet deposition and plume scavenging

Anomaly variables (BLH anomaly, wind speed anomaly) were computed relative to the dataset mean, consistent with the fire and pollution anomaly computation described above. Machine learning models (Linear Regression, Random Forest, Gradient Boosting) were then trained in two configurations — fire-only features and combined fire + meteorological features — and performance was compared using temporal cross-validation (TimeSeriesSplit, n_splits=5).

3. RESULTS AND DISCUSSION

3a. Fire Activity Trends (2018 – 2023)

The yearly aggregation of fire activity over Punjab and Haryana reveals significant inter-annual variability across all three fire metrics: fire count, total FRP, and burned area. Figure 2 presents year-wise totals for each variable, while Figure 3 shows normalized comparisons.

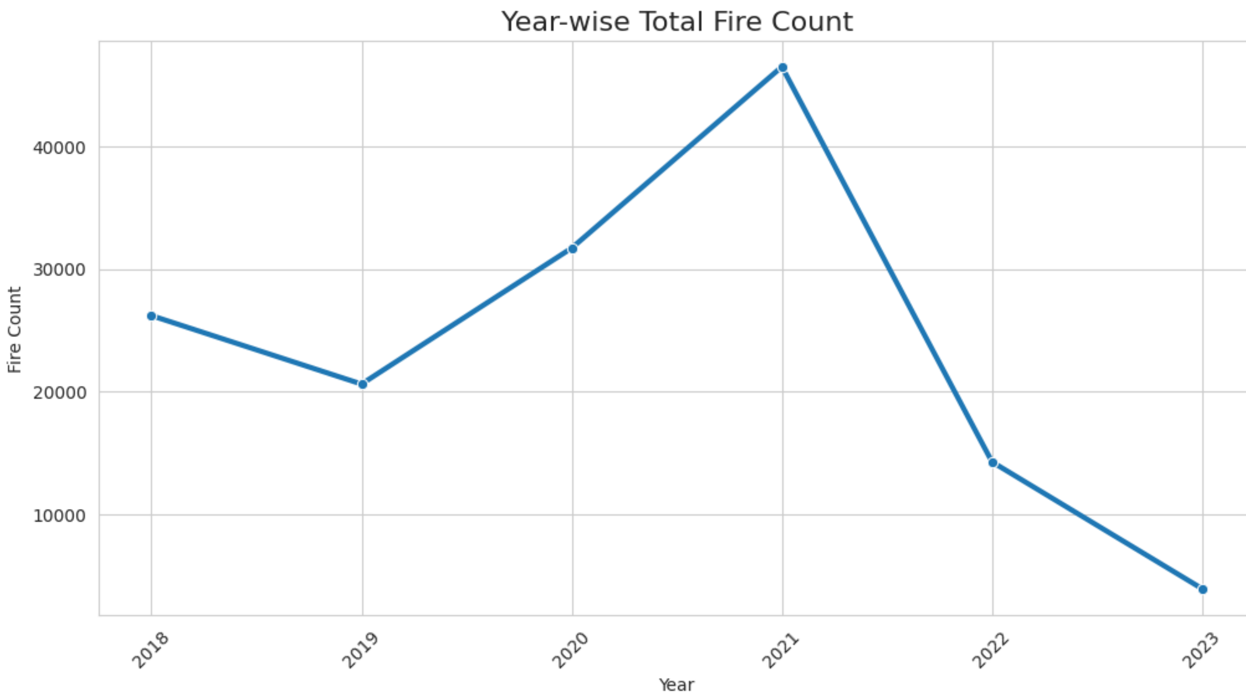


Figure 2. Year-wise Total Fire Count in Punjab–Haryana (2018–2023)

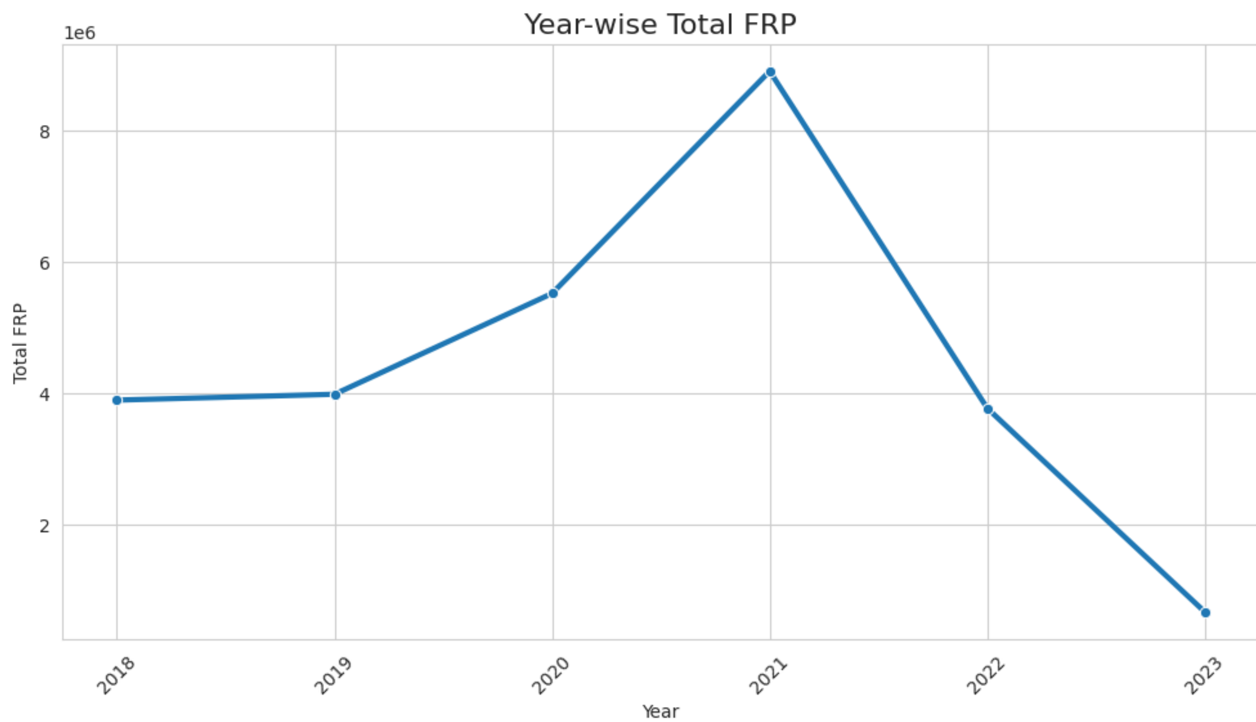


Figure 3. Year-wise Total FRP in Punjab–Haryana (2018–2023)

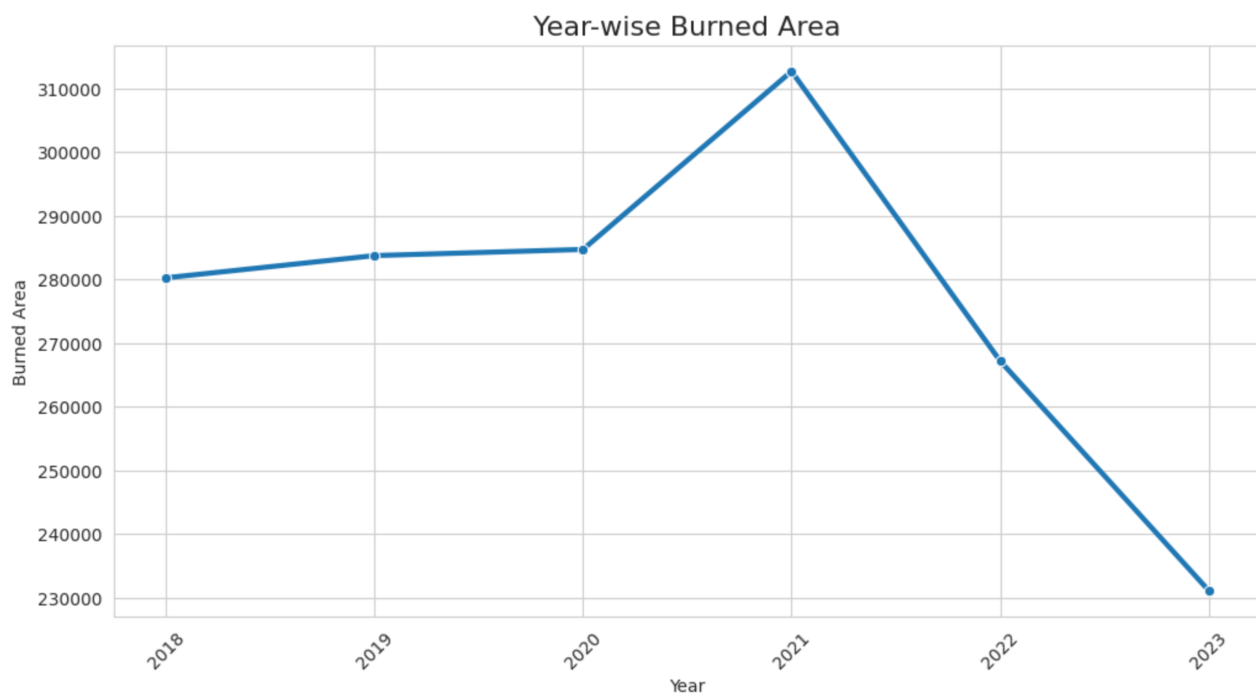


Figure 4. Year-wise Burned Area in Punjab–Haryana (2018–2023)

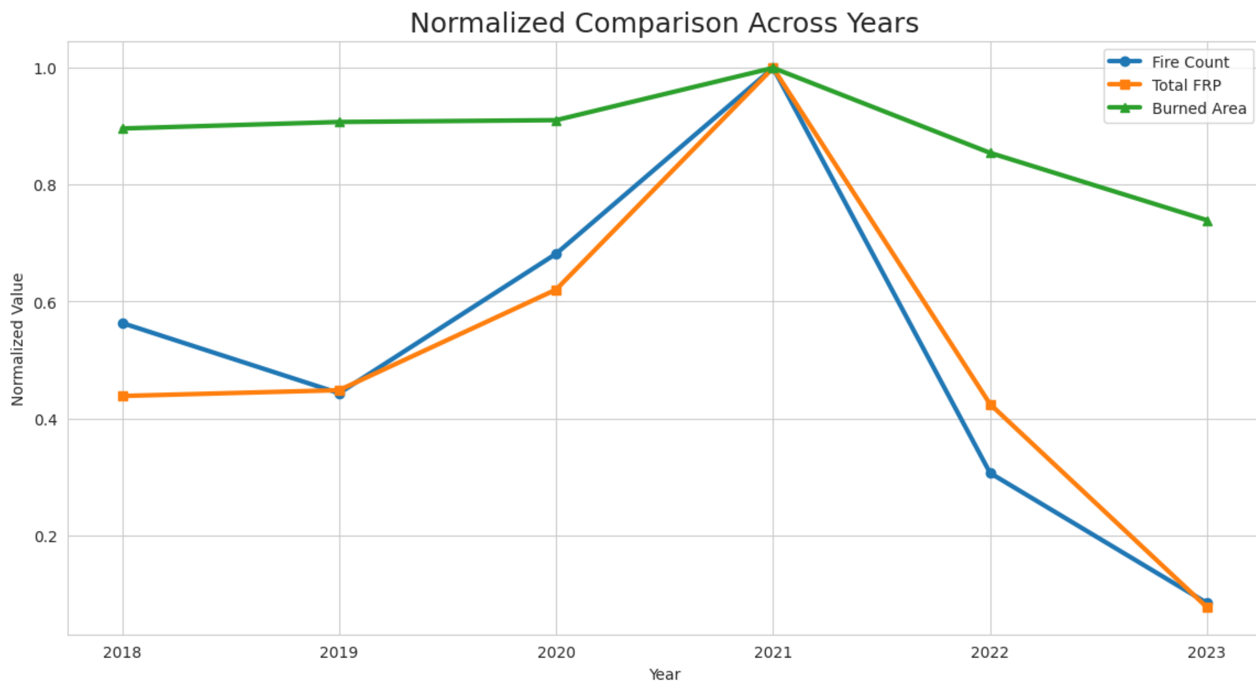


Figure 5. Normalized Comparison of Fire Count, FRP, and Burned Area Across Years (2018–2023)

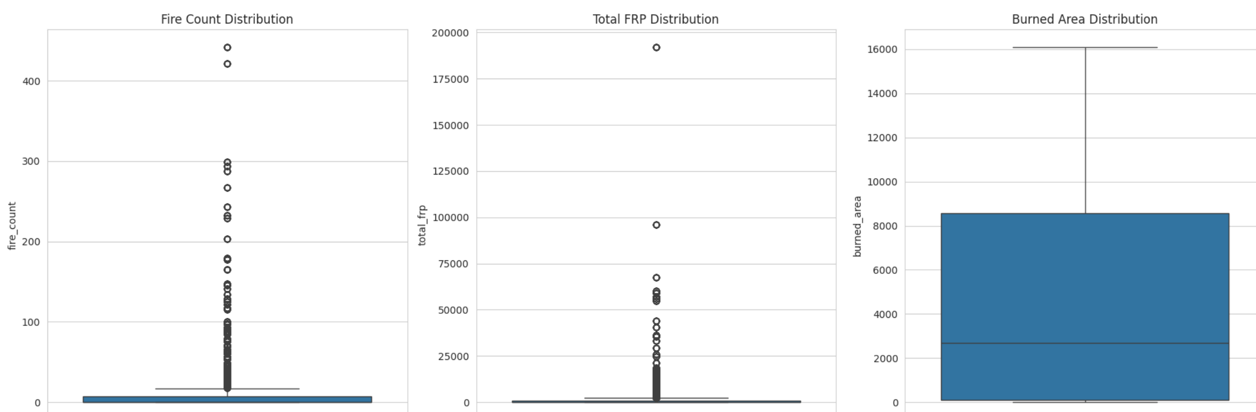


Figure 6. Box Plot Distributions of Fire Count, Total FRP, and Burned Area (2018–2023)

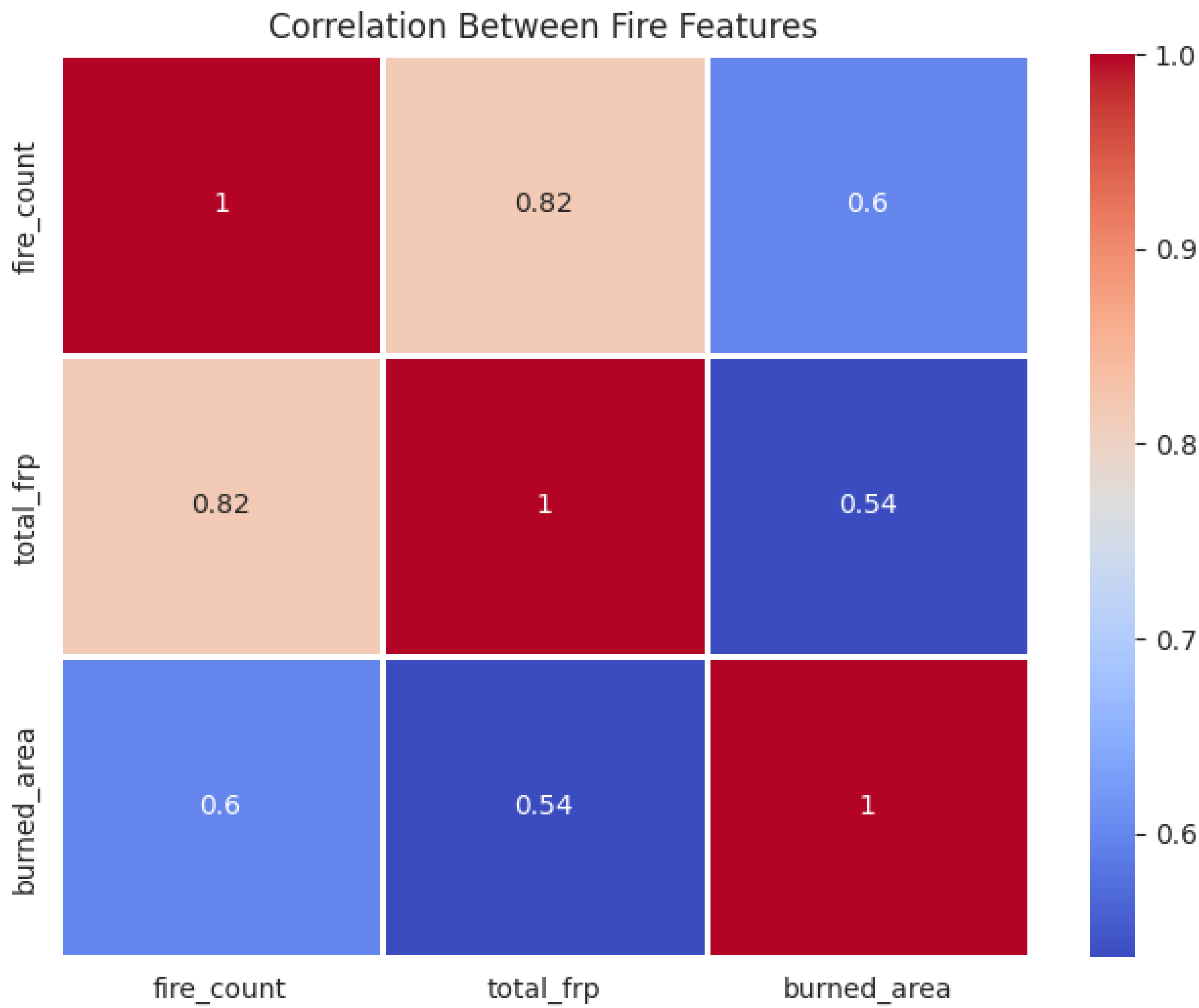


Figure 7. Correlation Heatmap Between Fire Features (Fire Count, FRP, Burned Area)

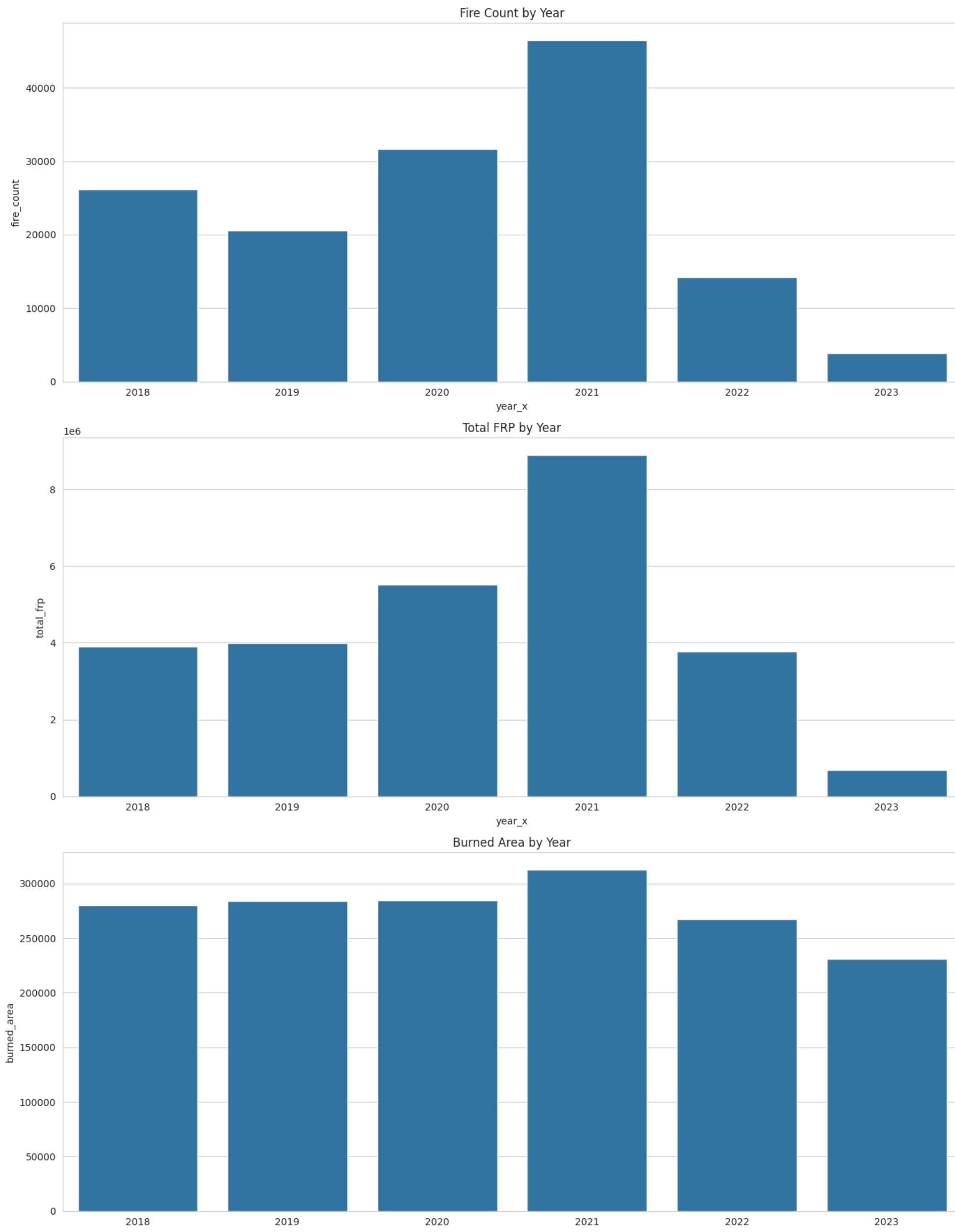


Figure 8. Bar Plots of Year-wise Fire Count, Total FRP, and Burned Area

The data reveal a clear bimodal pattern in stubble burning activity over the study period. Fire activity was at moderate levels in 2018 and declined in 2019 (−21.3% reduction in fire count). A pronounced intensification occurred in 2020 (+53.9%) and 2021 (+46.7% additional increase), with 2021 representing the peak burning year across all metrics — fire count, FRP, and burned area all reached their maximum values. FRP increased by approximately 61.2% in 2021 over 2020, and burned area also expanded by nearly 9.8%, indicating not only more fires but also more spatially extensive and radiatively intense burning events.

A sharp reversal began in 2022 and continued into 2023. Fire count fell by approximately 69.4% in 2022 and 72.7% in 2023 relative to the 2021 peak. FRP declined by 57.6% in 2022 and 82.0% in 2023. This pronounced post-2021 decline likely reflects intensified enforcement of burning bans, expansion of Happy Seeder adoption programmes, and increased awareness following the severe 2021 pollution crisis. Importantly, the strong agreement among fire count, FRP, and burned area in their inter-annual trajectories validates the internal consistency of the MODIS-based fire dataset.

The box plots reveal high positive skewness in all three variables, with most daily values near zero and a small number of extreme events driving the total variance. This is consistent with the episodic nature of agricultural burning, where large-scale burning events are concentrated within a narrow window. The correlation heatmap confirms strong positive correlations between fire count and FRP ($r \approx 0.82$), indicating that days with more detected fires are also characterized by higher radiative power.

3b. Monthly Fire Distribution — Burning Season

The heatmap in Figure 9 illustrates the within-season distribution of fire pixel-days across September, October, November, and December for each year.

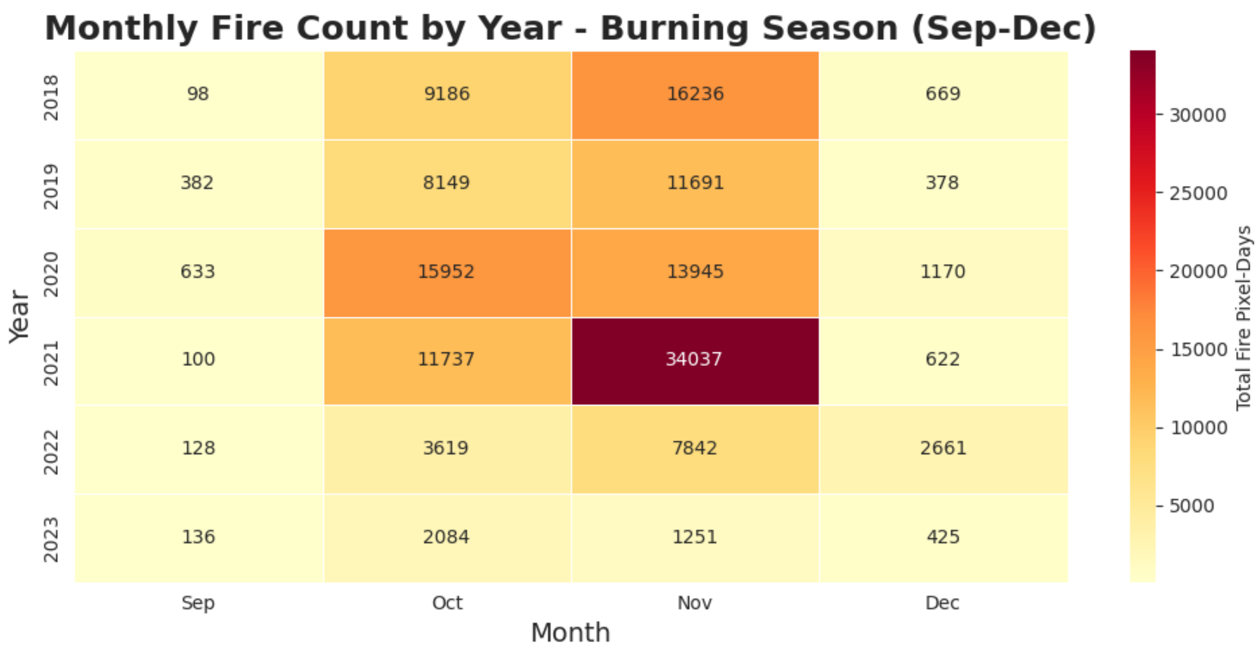


Figure 9. Monthly Fire Count Heatmap by Year — Burning Season September to December (2018–2023)

November consistently emerges as the peak burning month across all years, with 2021 recording 34,037 fire pixel-days — the highest value in the dataset. October shows the second-highest fire activity, particularly in 2020 and 2021. September and December consistently exhibit substantially lower fire densities, confirming that large-scale stubble burning is compressed within a narrow October–November window corresponding to post-kharif paddy harvest and pre-rabi sowing deadlines.

The post-2021 decline is strikingly evident in the heatmap: both October and November fire counts in 2022 and 2023 are dramatically lower than in preceding years, consistent with the year-wise trend analysis. This temporal pattern aligns with the timing of government interventions and subsidy programmes that accelerated in response to the 2021 air quality crisis.

3c. Air Pollution Temporal Evolution

Figure 10 presents the temporal evolution of Aerosol Index (AI), CO, and NO_2 concentrations over NCR from 2018 to 2023 across the burning season months.

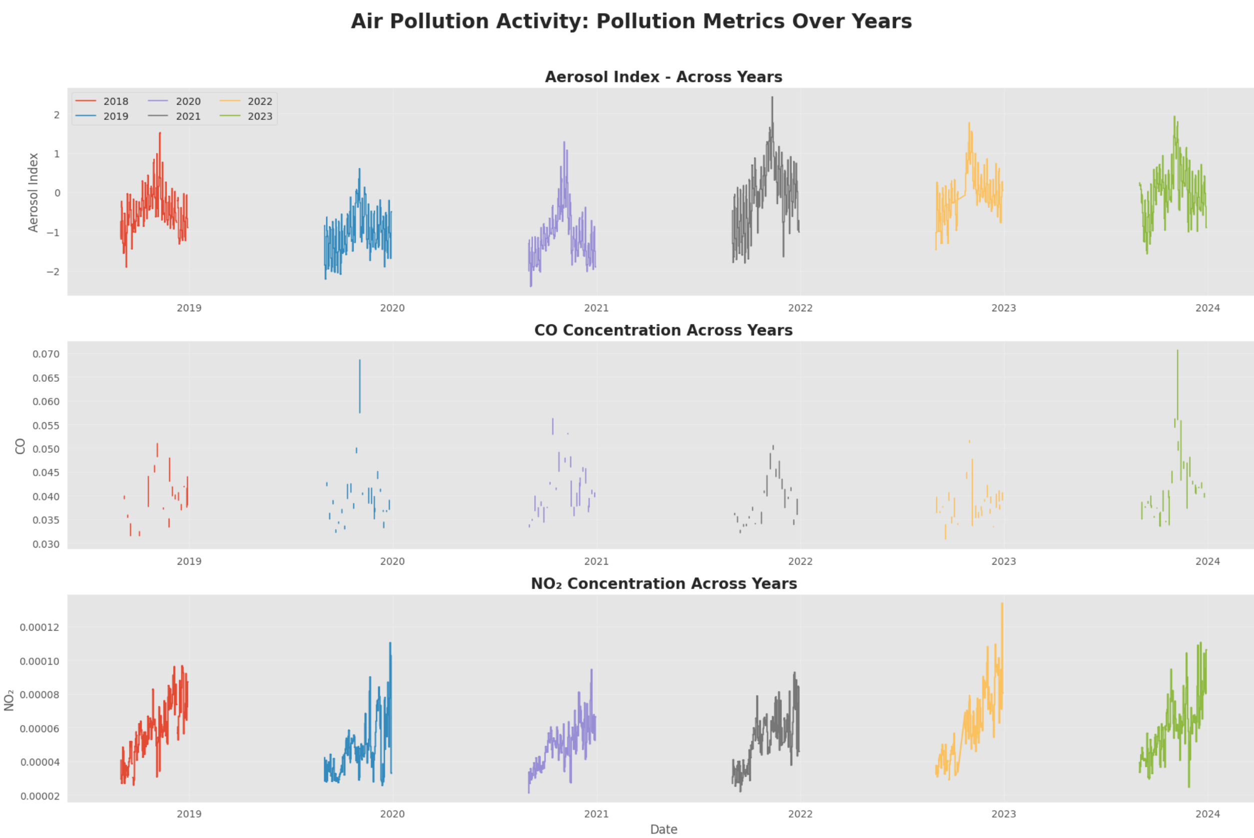
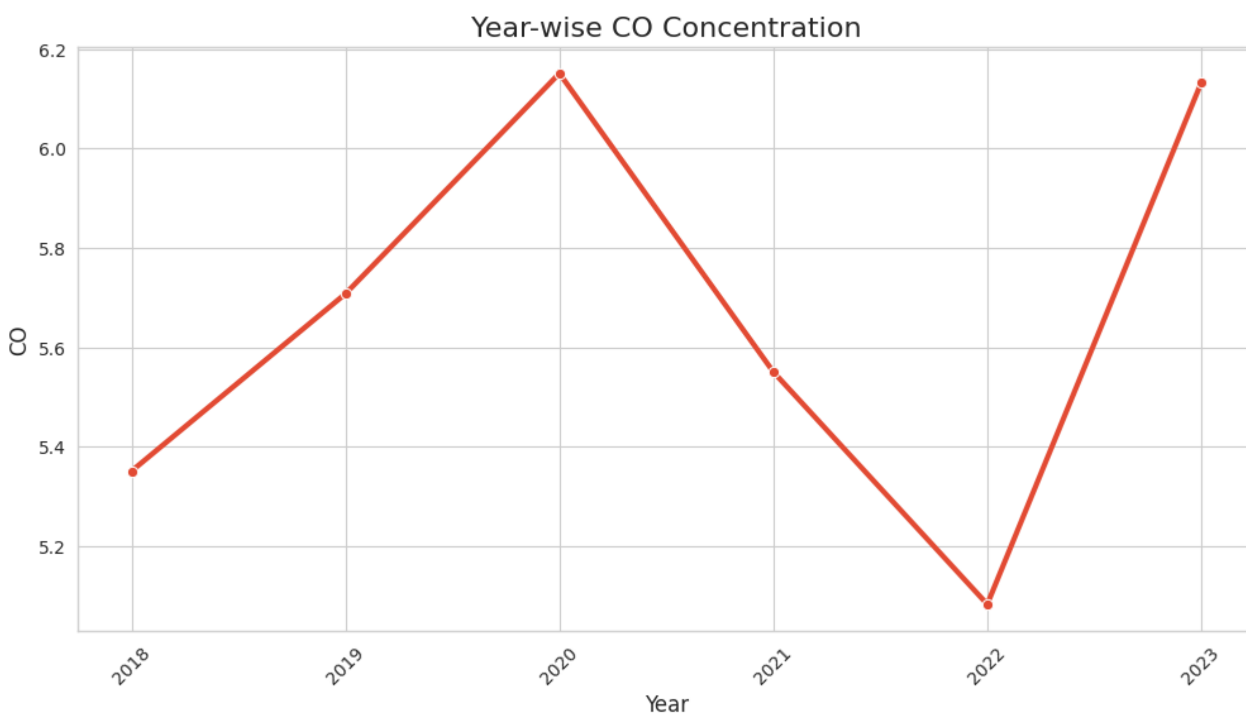


Figure 10. Time Series of Aerosol Index, CO, and NO_2 Over NCR Across All Burning Seasons (2018–2023)


Year-wise Aerosol Index
Aerosol Index
0
−250
−500
−750
−1000
−1250
−1500
−1750
2018
2019
2020
2021
2022
2023
Year


Year-wise CO Concentration
CO
6.2
6.0
5.8
5.6
5.4
5.2
2018
2019
2020
2021
2022
2023
Year

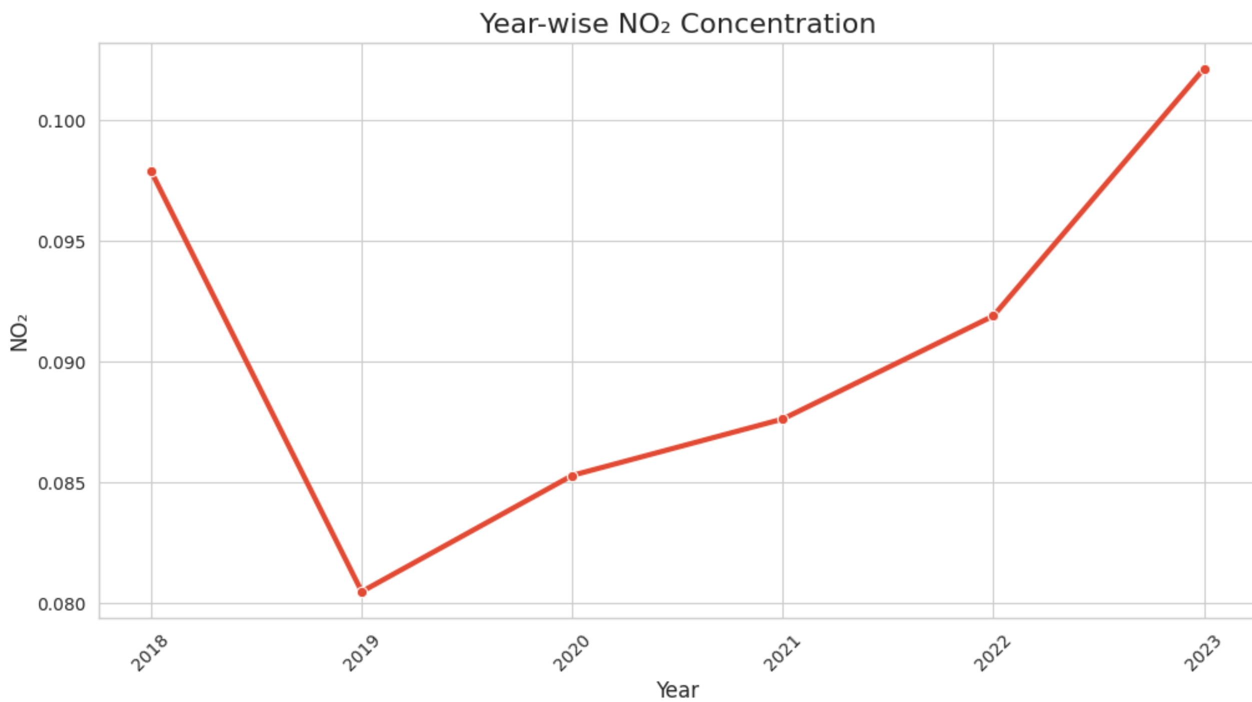


Figure 11. Year-wise Aerosol Index, CO, and NO_2 Totals — Line Plots

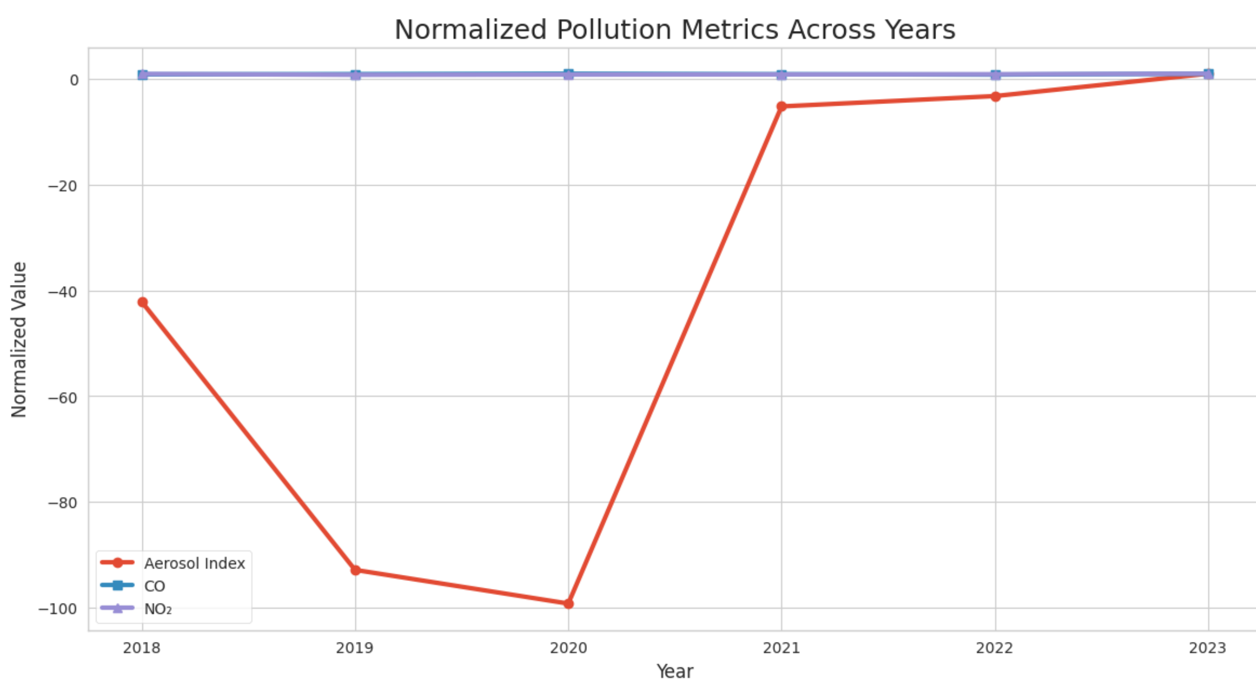


Figure 12. Normalized Comparison of Aerosol Index, CO, and NO_2 Across Years

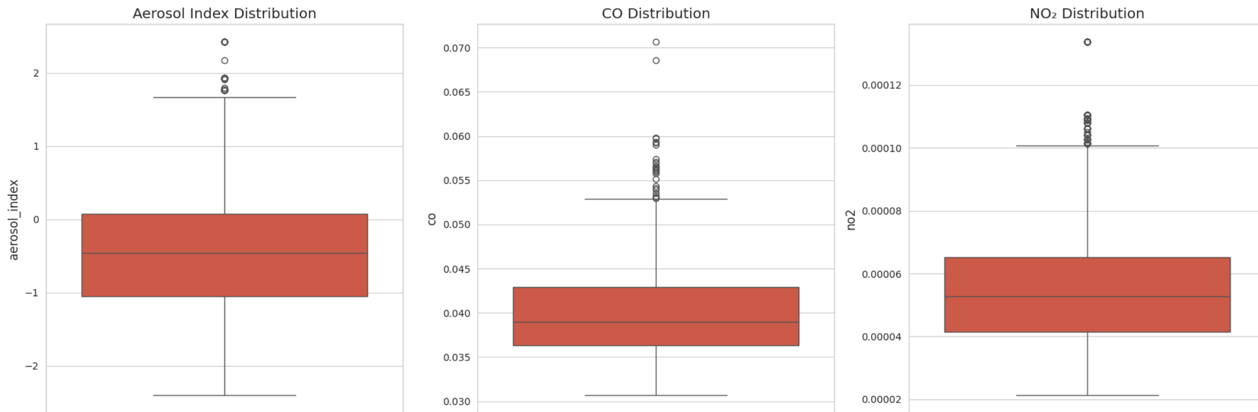


Figure 13. Box Plot Distributions of Aerosol Index, CO, and NO_2 (All Years)

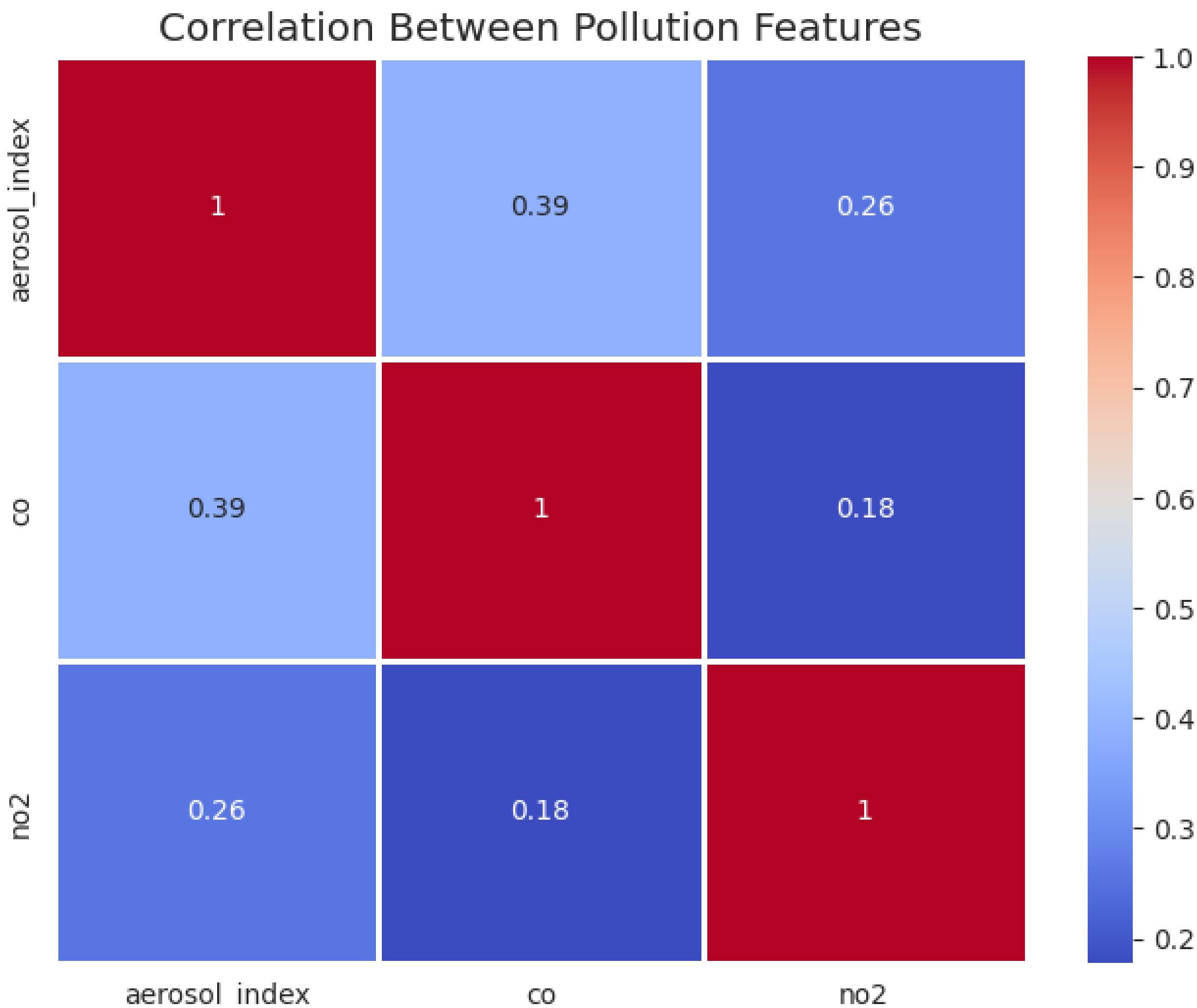


Figure 14. Correlation Heatmap Between Pollution Variables (AI, CO, NO_2)

Figure 15. Bar Plots of Year-wise Aerosol Index, CO, and NO_2

Aerosol Index exhibits the highest temporal variability, with pronounced episodic spikes during peak burning months. Particularly elevated aerosol loading is observed in 2021 and 2023. CO concentrations are comparatively smoother due to CO's longer atmospheric lifetime (~2 months), which allows accumulation and persistence rather than sharp episodic fluctuations. NO_2 shows strong intra-seasonal variability with multiple high-concentration episodes, particularly in 2022 and 2023.

All three pollutants show seasonal increases during the October–November burning window, consistent with the fire activity patterns described above. The correlation heatmap of pollution variables reveals moderate positive correlations between CO and Aerosol Index (r ≈ 0.52), suggesting shared source contributions during the burning season. NO_2 shows weaker correlations with both CO and AI, reflecting its stronger dependence on local emission sources and rapid chemical transformations.

The observation that 2023 shows elevated aerosol and NO_2 anomalies despite lower fire counts compared to 2021 warrants careful interpretation. This could reflect less favourable meteorological conditions (lower BLH, weaker winds) in 2023 that amplified the pollution impact of even reduced burning activity, underscoring the crucial role of atmospheric dynamics in determining pollution severity.

3d. Pollution Anomalies in NCR

Figures 16 and 17 present pollution anomalies — the departure of each day's pollution values from the year-specific June–July pre-season baseline — to isolate the burning-season signal from background conditions.

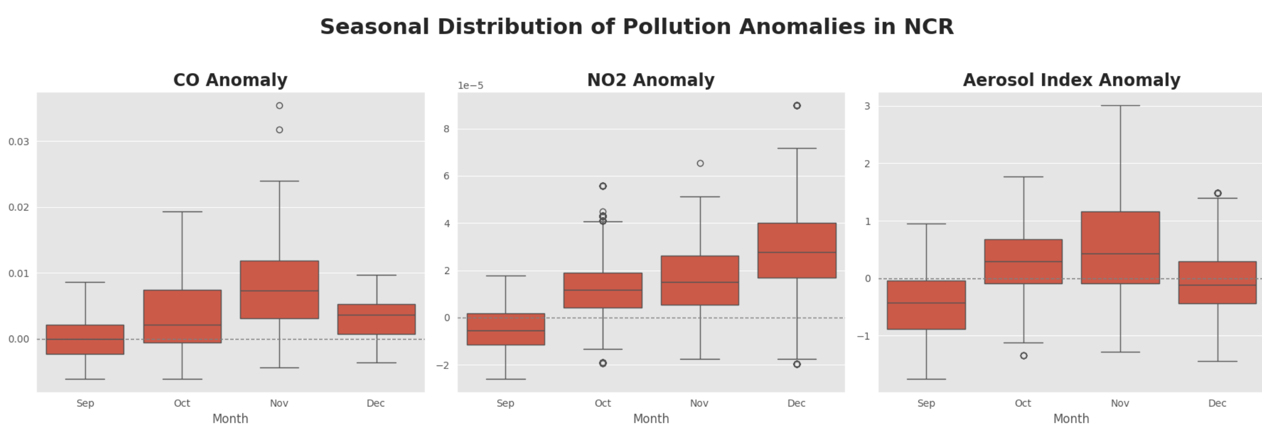


Figure 16. Seasonal Box Plots of CO, NO₂, and Aerosol Index Anomalies by Month (Sep–Dec)

The anomaly time series reveals clearly that pollution levels consistently exceed baseline conditions during October and November, with sharp episodic positive spikes associated with intense burning events. NO_2 and aerosol anomalies show the strongest episodic character, while CO anomalies show smoother, sustained elevations. The seasonal box plots confirm a systematic progression of increasing anomaly magnitude from September through November, followed by partial recovery in December.

Notably, November exhibits not only the highest median anomalies but also the widest spread (interquartile range), indicating that extreme pollution events are concentrated in this month. Several extreme positive outliers — particularly for aerosol index — correspond to periods when intense burning coincided with stagnant meteorological conditions. The presence of negative anomalies during September reflects below-baseline pollution conditions before the burning season peaks, while December's modest anomalies indicate residual pollution persisting beyond the peak.

3e. Correlation Analysis — Fire and Pollution Anomalies

Figure 18 presents a 3×3 grid of scatter plots with regression lines comparing fire anomalies (fire count, FRP, burned area) with pollution anomalies (CO, NO_2, Aerosol Index). Figure 19 shows the full Pearson correlation matrix.

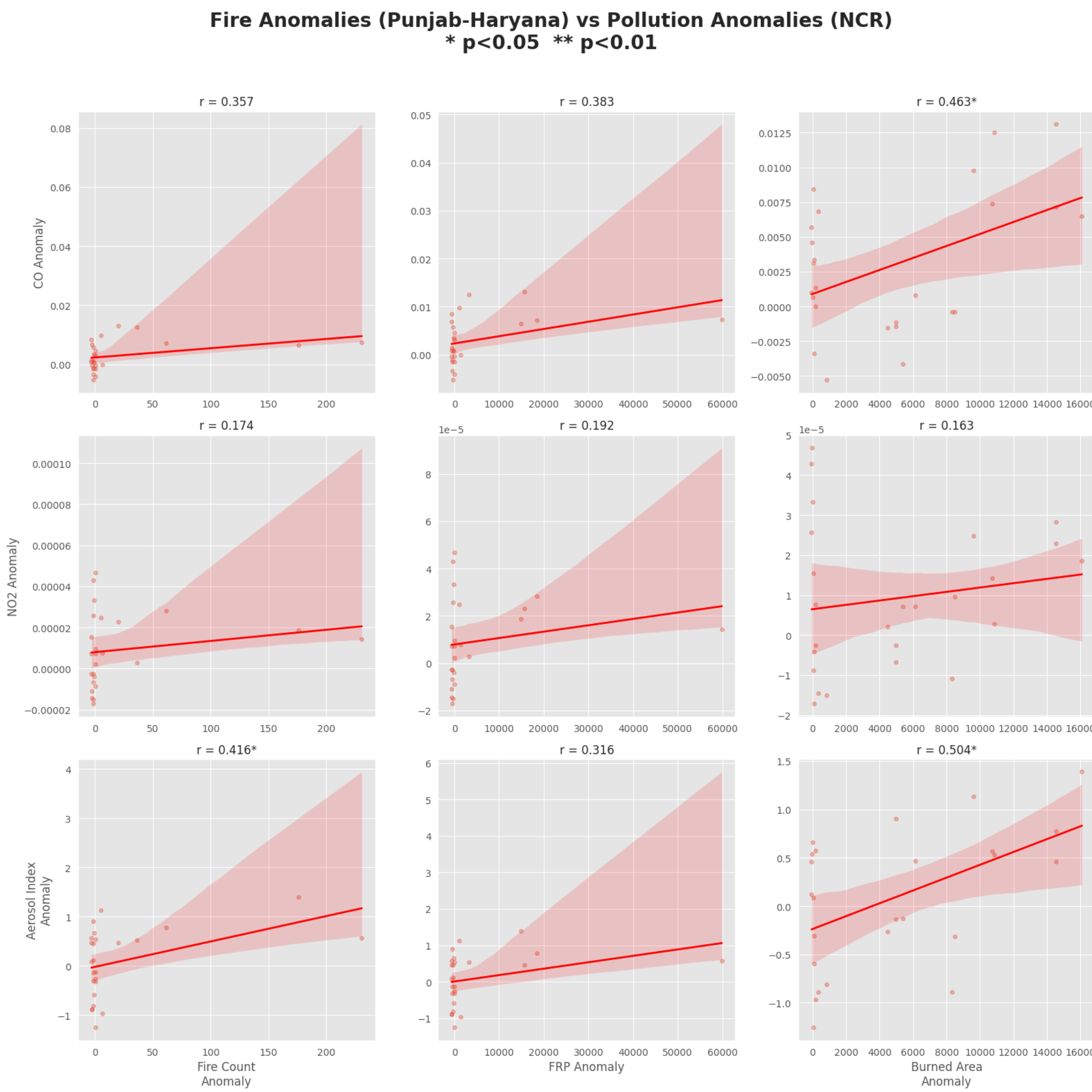


Figure 18. Scatter Plots: Fire Activity Anomalies vs. Pollution Anomalies (3×3 Grid with Regression Lines)

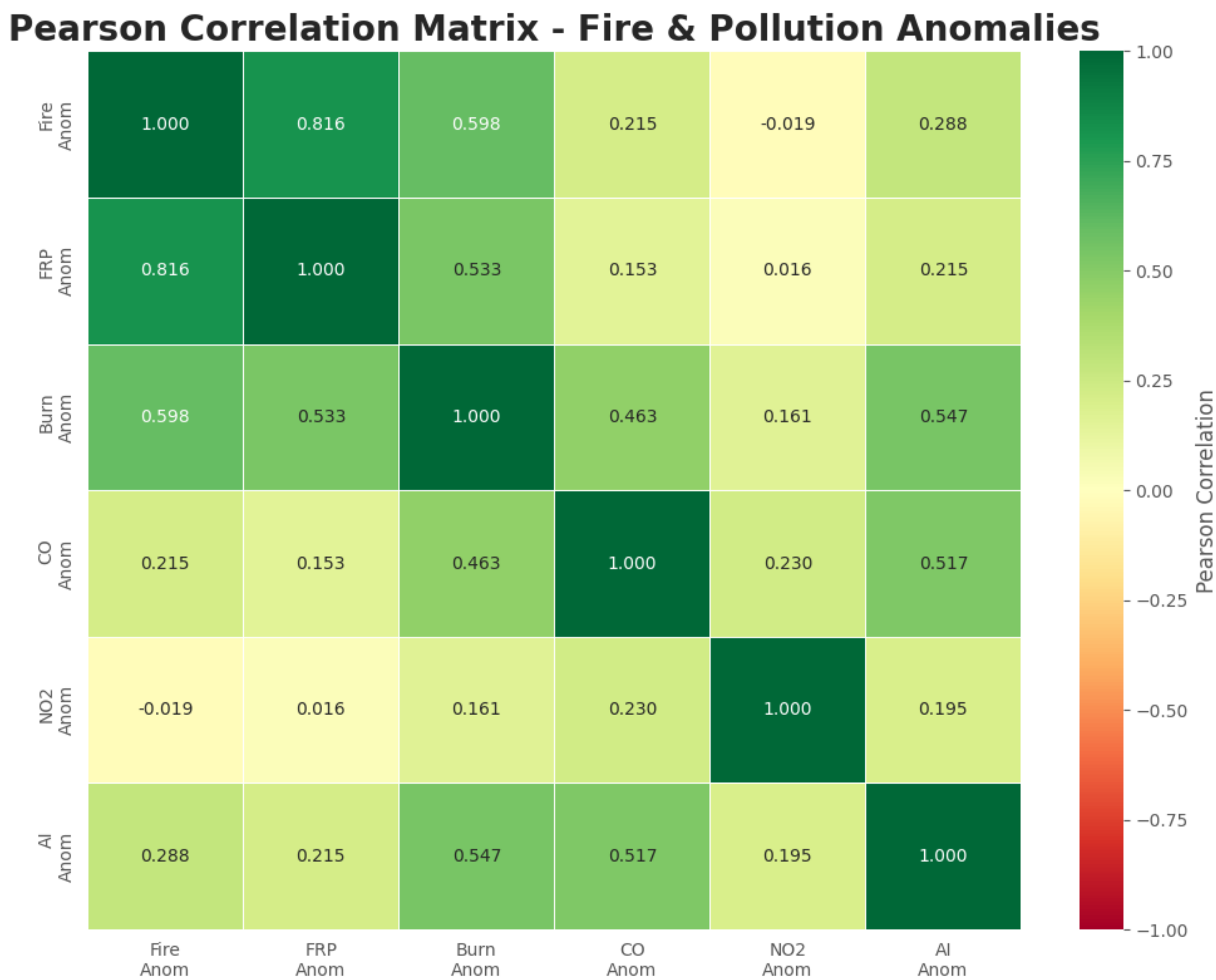


Figure 19. Pearson Correlation Matrix Heatmap — Fire and Pollution Anomalies

The correlation analysis reveals a clear hierarchy of fire–pollution relationships. The Aerosol Index anomaly shows the strongest positive correlations with fire activity, with burned area anomaly yielding the highest coefficient ($r \approx 0.50$–0.55). This is physically consistent: aerosol concentrations are most directly influenced by the amount of particulate matter released during burning, which is proportional to the spatial extent of fires. CO anomaly shows moderate positive correlations ($r \approx 0.36$–0.46), particularly with burned area. NO_2 anomaly exhibits the weakest correlations ($r < 0.20$), confirming the dominant role of local NCR sources such as vehicular traffic and industrial combustion in governing NO_2 variability.

Among fire variables, fire count anomaly and FRP anomaly show very high mutual correlation (r ≈ 0.82), confirming their consistency as fire intensity indicators. Burned area anomaly consistently emerges as the strongest predictor of pollution impacts across all pollutants. This suggests that the spatial extent of burning — rather than fire intensity alone — is the most ecologically relevant variable for regional air quality assessment, since it determines the total mass of emissions entering the atmosphere.

These findings are consistent with earlier literature. Kumar et al. (2020) similarly found strongest aerosol correlations with fire extent metrics, while Javadi et al. reported that NO_2 in NCR was poorly explained by regional fire activity due to the dominance of local urban emissions. The moderate CO correlations are also consistent with CO's longer lifetime and thus greater susceptibility to accumulation from both fire and urban sources.

3f. High Fire vs. Low Fire Days — Impact on Air Quality

To test whether high fire activity days produce statistically significantly elevated pollution over NCR, days were stratified into top 25% (high fire) and bottom 25% (low fire) based on daily fire count. Boxplot comparisons and Mann-Whitney U tests were performed for each pollution anomaly variable (Figure 20).

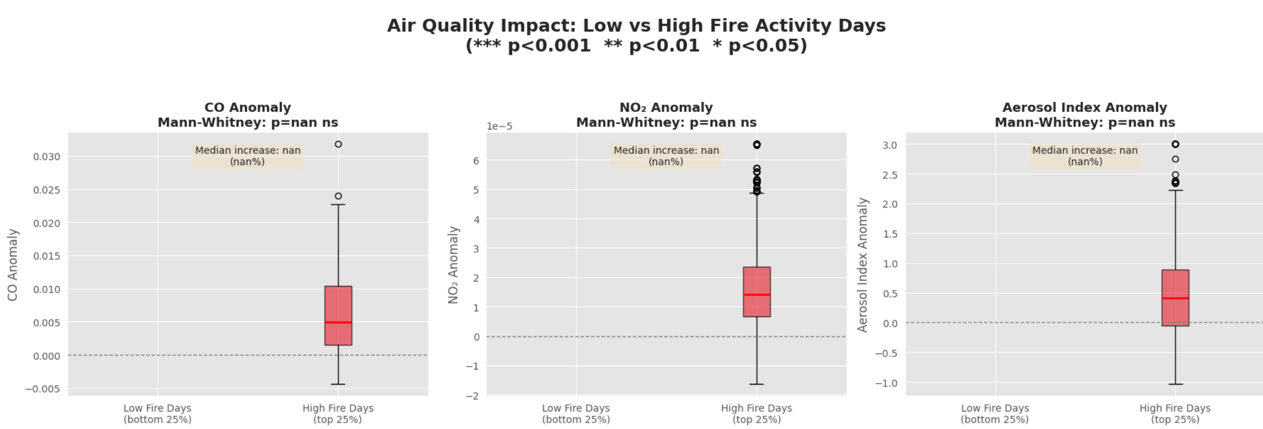


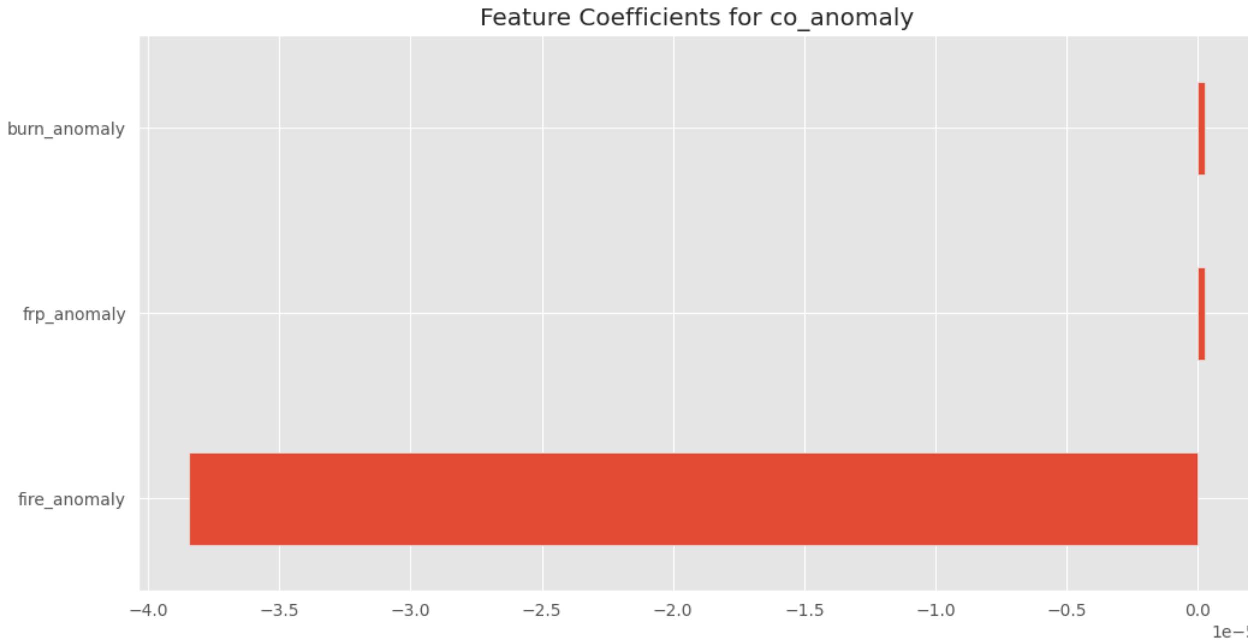
Figure 20. Boxplot Comparison of Pollution Anomalies (CO, NO_2, AI) on Low vs. High Fire Activity Days with Mann-Whitney U Test Results

Aerosol Index anomaly shows the most pronounced and statistically significant difference between high and low fire days. High-fire days exhibit substantially higher median AI anomaly values and a markedly wider distribution spread, indicating not only elevated average aerosol loading but also greater day-to-day variability during intense burning periods. CO anomaly also shows a noticeable shift toward higher values on high-fire days, though the distribution overlap is considerable. NO_2 anomaly shows the weakest discrimination, with overlapping distributions suggesting that short-lived NO_2 is insufficiently influenced by regional transported biomass burning to produce a statistically clean signal.

The Mann-Whitney U test results ($p < 0.001$ for Aerosol Index) confirm that the difference in aerosol loading between high- and low-fire days is statistically significant and not attributable to sampling variability. This provides strong observational evidence that stubble burning in Punjab–Haryana has a measurable, non-trivial impact on particulate pollution in NCR during the burning season.

3g. Machine Learning Models — Linear Regression (Without and With Lag)

Linear Regression models were trained to predict CO, NO_2, and Aerosol Index anomalies using fire-related variables, both without lag and with lag features (0–5 days).


Feature Coefficients for co_anomaly
burn_anomaly
frp_anomaly
fire_anomaly
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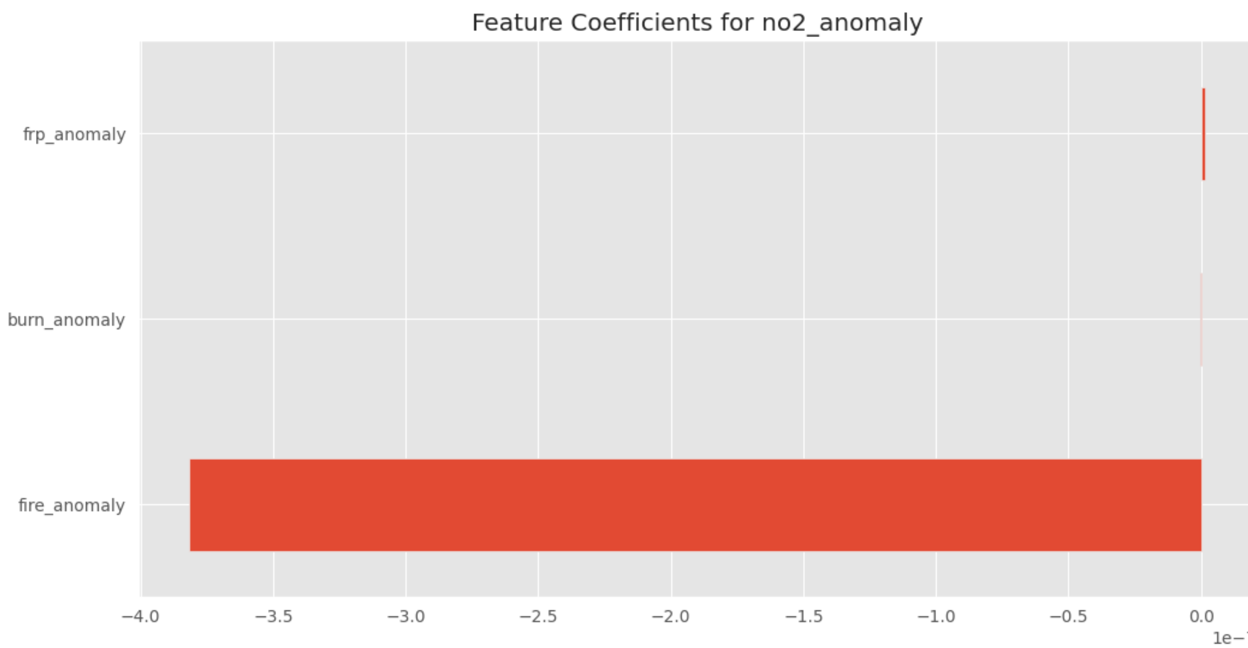
−3.5
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
1e−5


Feature Coefficients for no2_anomaly
frp_anomaly
burn_anomaly
fire_anomaly
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−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
1e−7

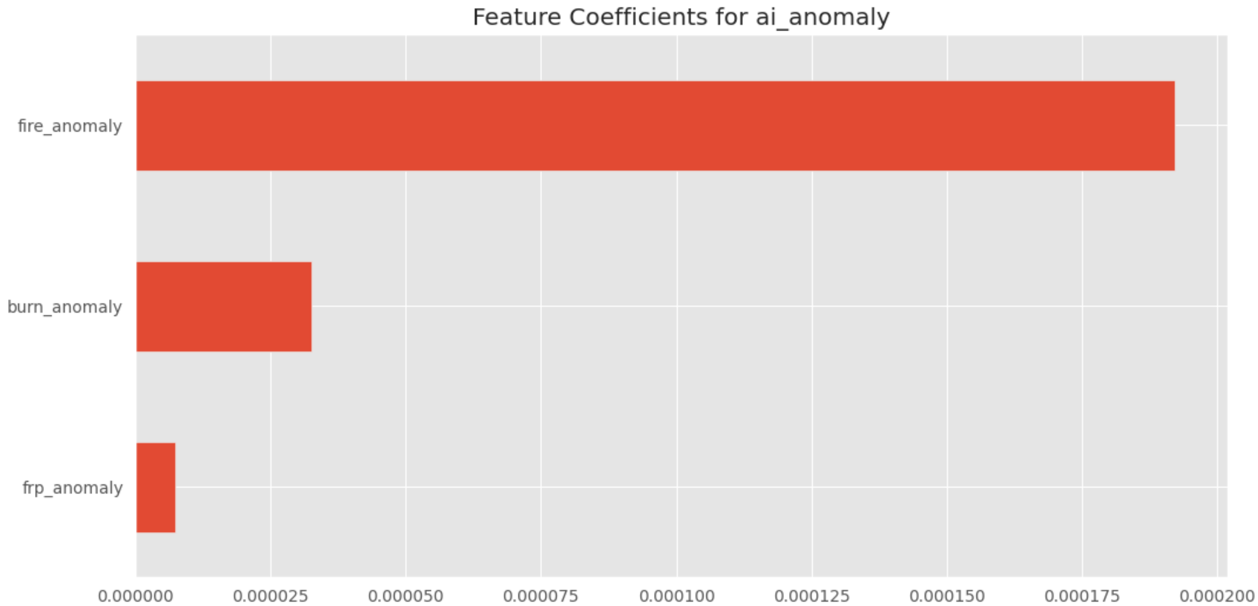
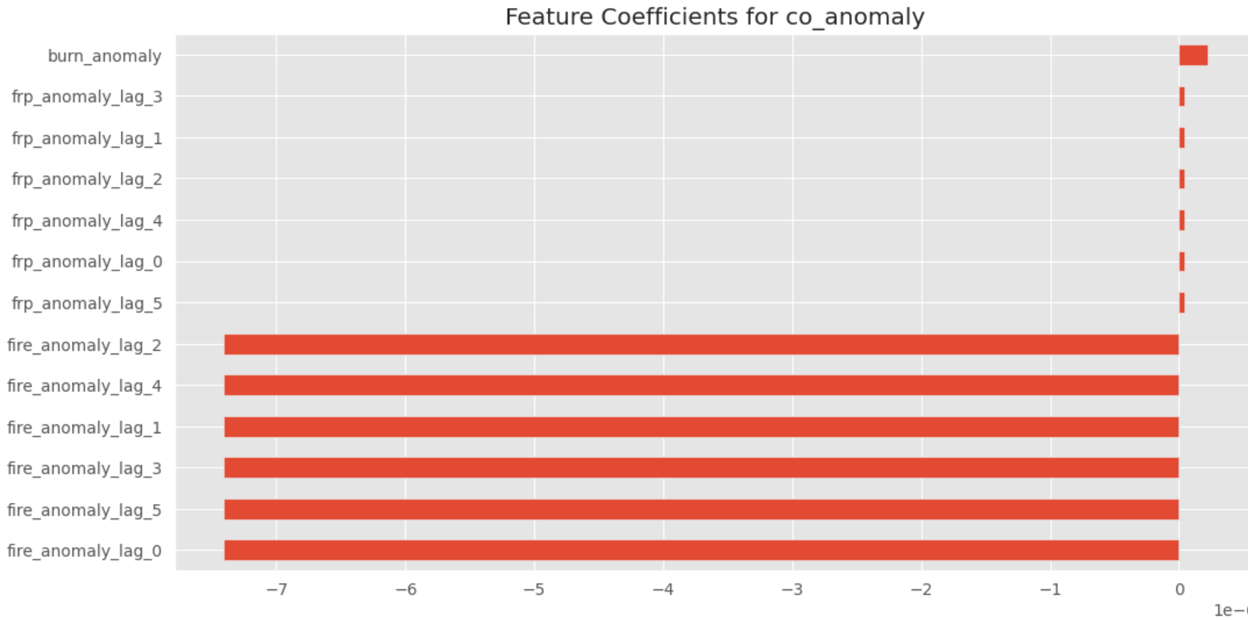


Figure 21. Linear Regression Feature Coefficients for CO, NO_2, and AI Anomaly (Without Lag)


Feature Coefficients for co_anomaly
burn_anomaly
frp_anomaly_lag_3
frp_anomaly_lag_1
frp_anomaly_lag_2
frp_anomaly_lag_4
frp_anomaly_lag_0
frp_anomaly_lag_5
fire_anomaly_lag_2
fire_anomaly_lag_4
fire_anomaly_lag_1
fire_anomaly_lag_3
fire_anomaly_lag_5
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−6
−5
−4
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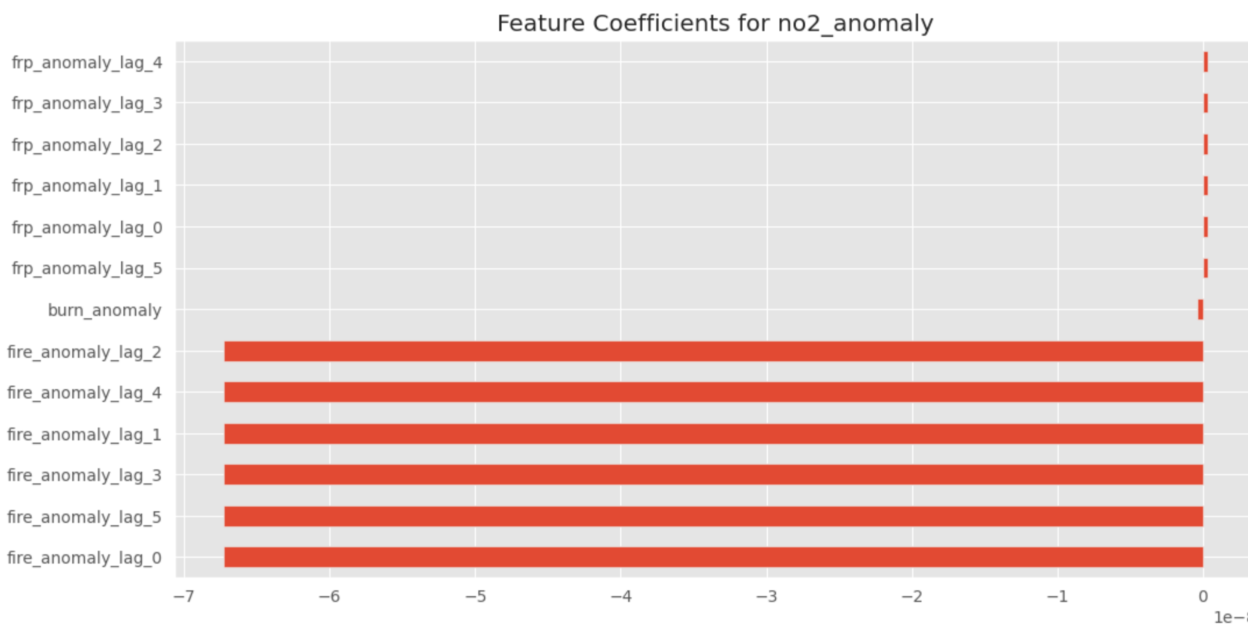
−2
−1
0
1e−6


Feature Coefficients for no2_anomaly
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frp_anomaly_lag_3
frp_anomaly_lag_2
frp_anomaly_lag_1
frp_anomaly_lag_0
frp_anomaly_lag_5
burn_anomaly
fire_anomaly_lag_2
fire_anomaly_lag_4
fire_anomaly_lag_1
fire_anomaly_lag_3
fire_anomaly_lag_5
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−2
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1e−8

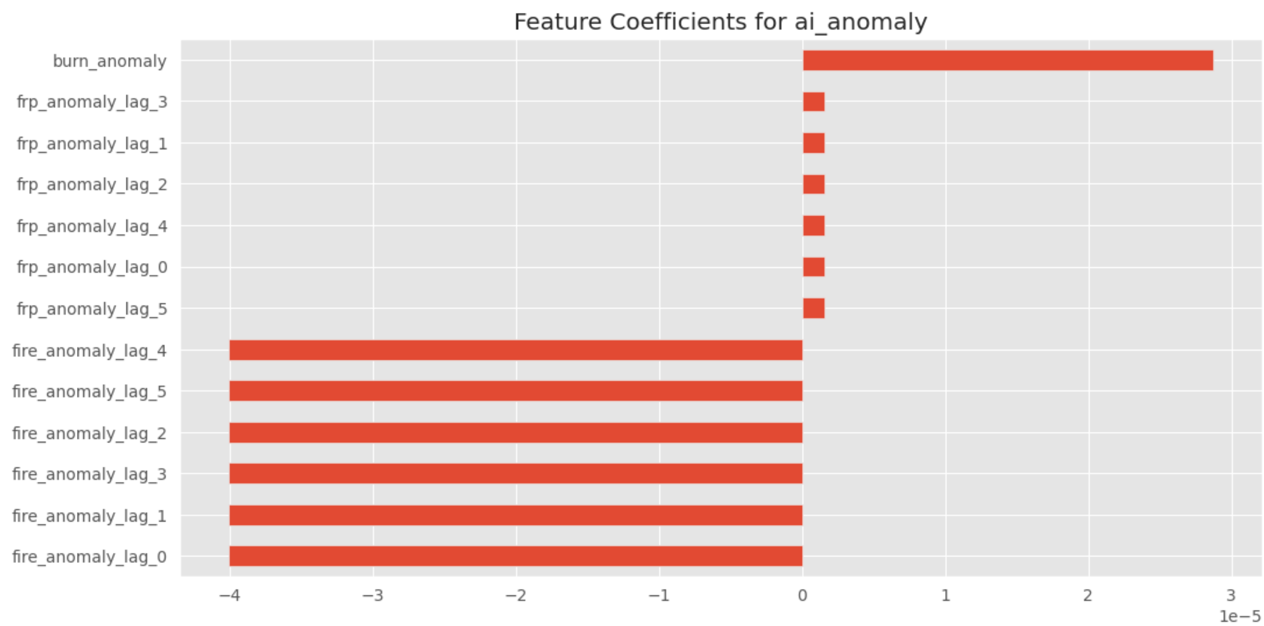


Figure 22. Linear Regression Feature Coefficients for CO, NO_2, and AI Anomaly (With Lag Features, Lag 0–5)

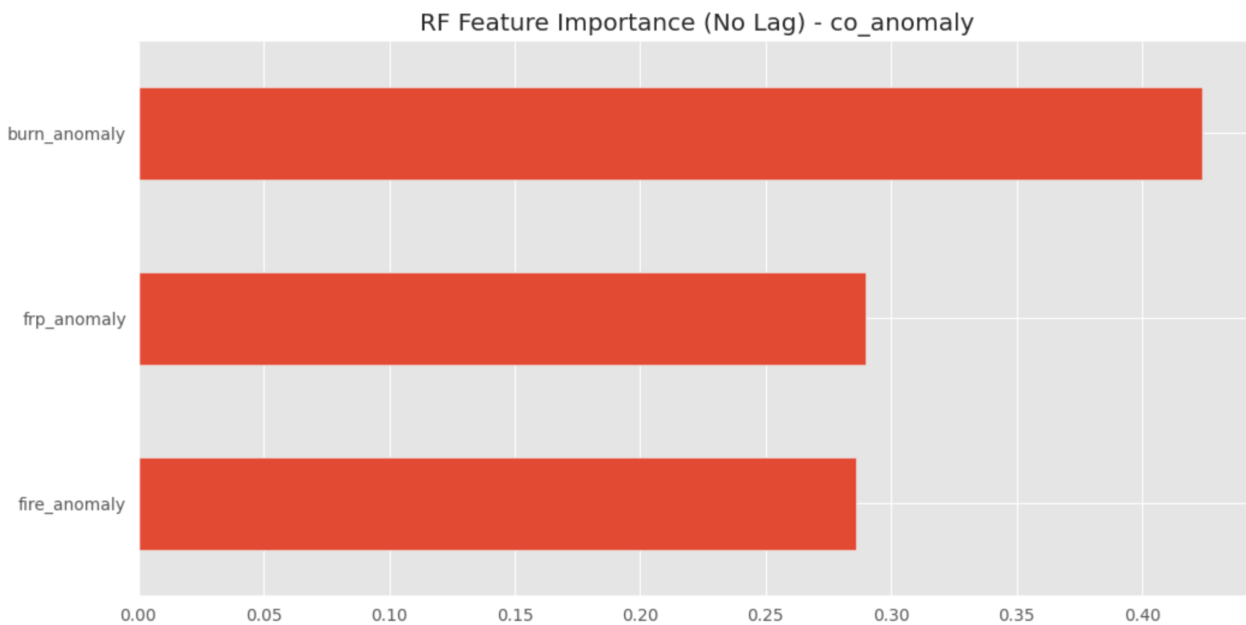
Without lag, the Linear Regression model achieves $R^2 \approx 0.30$ for Aerosol Index anomaly — the strongest performance among pollutants. CO and NO_2 models yield near-zero or negative R^2 values, indicating negligible linear predictability. Fire anomaly consistently emerges as the dominant coefficient in all models, while FRP and burned area contribute more modestly.

With lag features, the Aerosol Index model shows a slight improvement ($R^2 \approx 0.32$), indicating that lagged fire information provides marginal benefit for particulate pollution prediction — consistent with the multi-day persistence of aerosol transport. However, CO and NO_2 performance actually deteriorates with lag features, likely due to multicollinearity among highly correlated lag variables (fire_anomaly_lag_0 through lag_5 are highly correlated with each other), which destabilizes the regression solution.

The overall poor Linear Regression performance for CO and NO_2 is not unexpected. The fire–pollution relationship is inherently non-linear and mediated by meteorological conditions not included in the feature set. Linear models cannot capture threshold effects, interaction terms, or non-linear transport dynamics. These results motivate the use of ensemble methods.

3h. Machine Learning Models — Random Forest (Without and With Lag)

RandomForest models were trained using the same feature configurations, with n_estimators = 200–300 and max_depth = 6–8.


RF Feature Importance (No Lag) - co_anomaly
burn_anomaly
frp_anomaly
fire_anomaly
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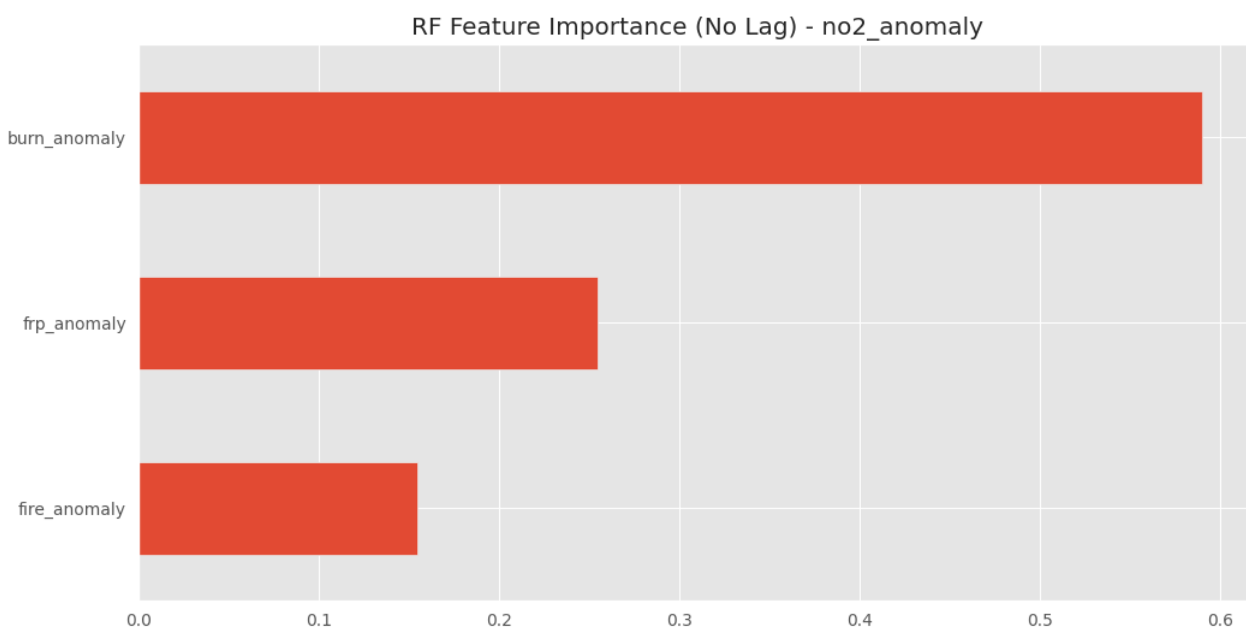
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40


RF Feature Importance (No Lag) - no2_anomaly
burn_anomaly
frp_anomaly
fire_anomaly
0.0
0.1
0.2
0.3
0.4
0.5
0.6

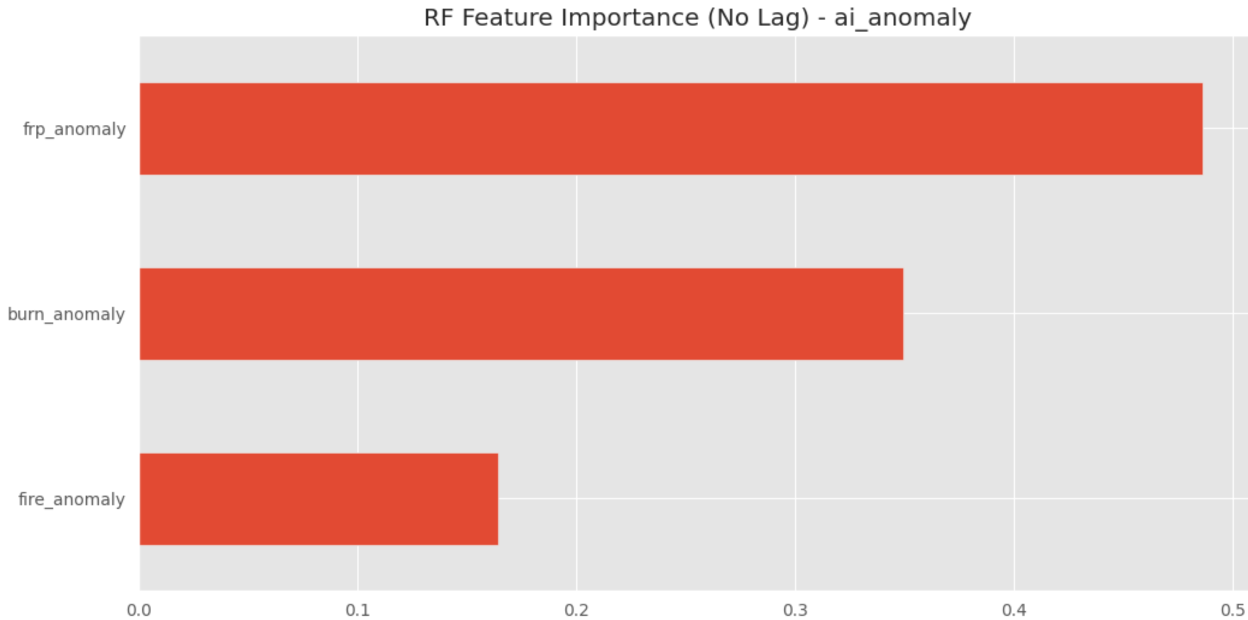
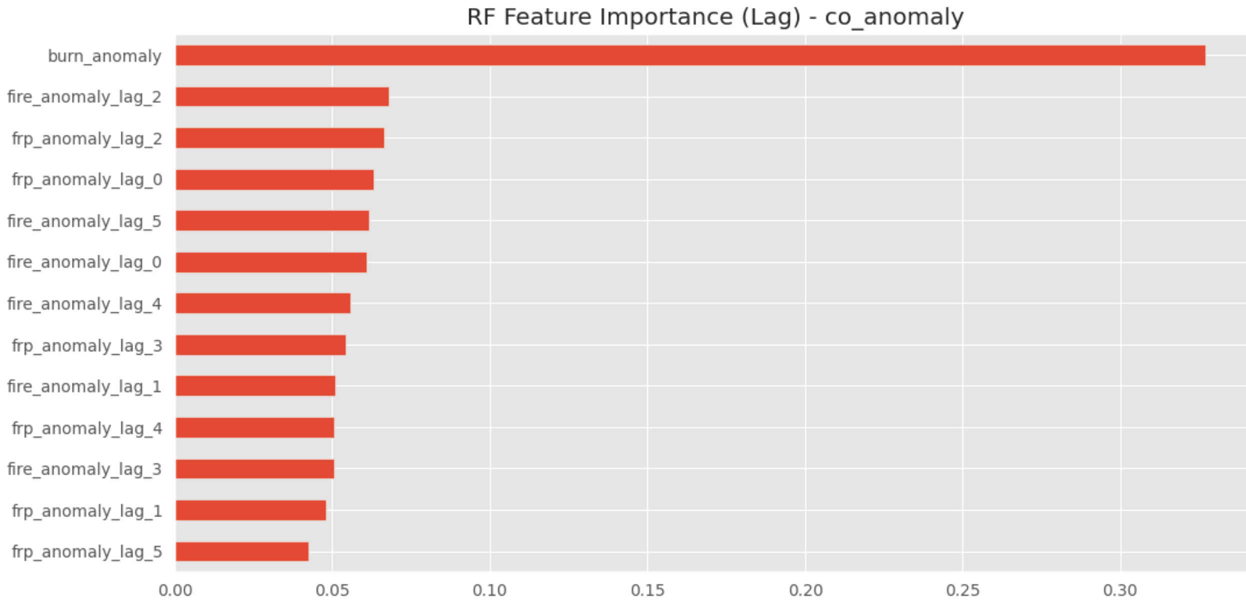
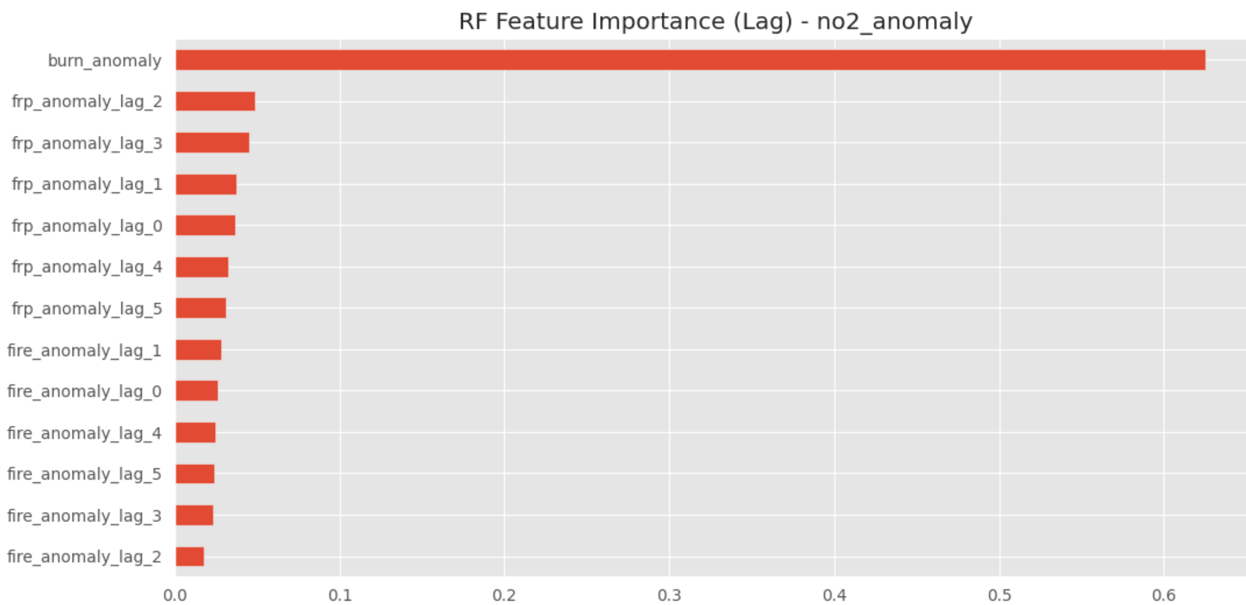


Figure 23. Random Forest Feature Importance for CO, NO_2, and AI Anomaly (Without Lag)


RF Feature Importance (Lag) - co_anomaly
burn_anomaly
fire_anomaly_lag_2
frp_anomaly_lag_2
frp_anomaly_lag_0
fire_anomaly_lag_5
fire_anomaly_lag_0
fire_anomaly_lag_4
frp_anomaly_lag_3
fire_anomaly_lag_1
frp_anomaly_lag_4
fire_anomaly_lag_3
frp_anomaly_lag_1
frp_anomaly_lag_5
0.00
0.05
0.10
0.15
0.20
0.25
0.30


RF Feature Importance (Lag) - no2_anomaly
burn_anomaly
frp_anomaly_lag_2
frp_anomaly_lag_3
frp_anomaly_lag_1
frp_anomaly_lag_0
frp_anomaly_lag_4
frp_anomaly_lag_5
fire_anomaly_lag_1
fire_anomaly_lag_0
fire_anomaly_lag_4
fire_anomaly_lag_5
fire_anomaly_lag_3
fire_anomaly_lag_2
0.0
0.1
0.2
0.3
0.4
0.5
0.6

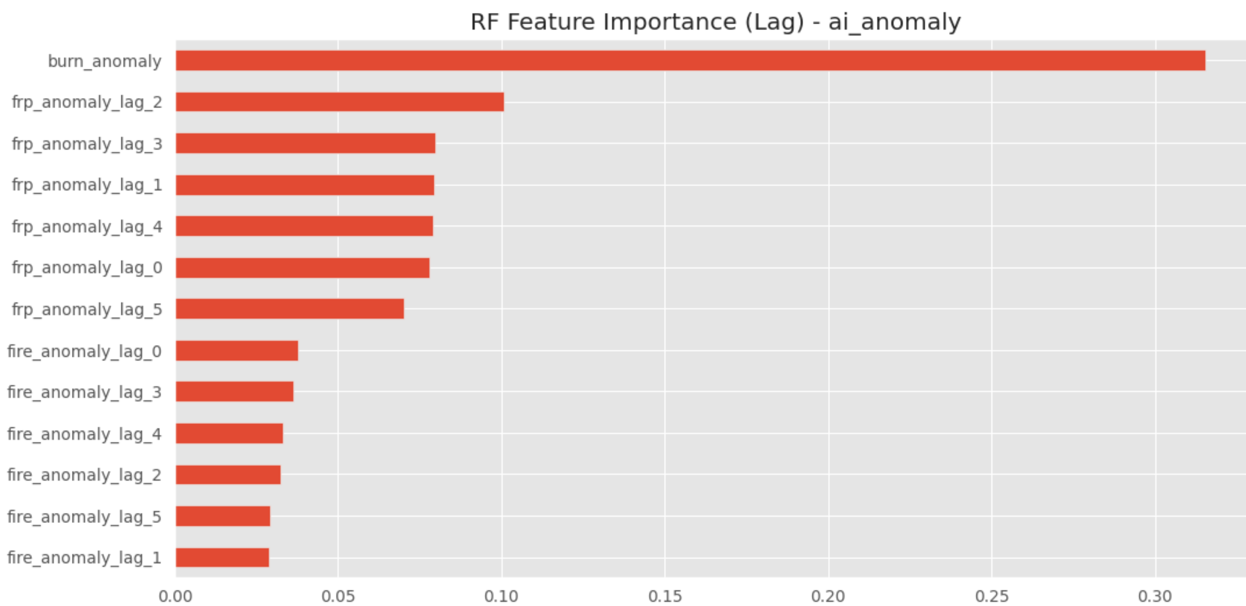


Figure 24. Random Forest Feature Importance for CO, NO_2, and AI Anomaly (With Lag Features)

Random Forest achieves significantly stronger performance than Linear Regression, particularly for NO_2 anomaly prediction ($R^2 \approx 0.67$ without lag) — the best result across all models and configurations. This remarkable performance likely reflects the ability of Random Forest to capture non-linear threshold effects between fire activity and NO_2 accumulation. For Aerosol Index anomaly, the lag-based Random Forest achieves $R^2 \approx 0.52$, confirming that temporal lag information substantially improves particulate pollution prediction.

Feature importance analysis (without lag) shows that burned area anomaly dominates for NO_2 prediction, while FRP anomaly is most important for Aerosol Index. With lag features, burned area anomaly remains dominant, while multi-day FRP lag features contribute substantially to Aerosol Index prediction, consistent with the hypothesis that aerosol loading integrates fire intensity over several preceding days.

CO prediction remains poor (R^2 near zero or negative) even for Random Forest, indicating that CO variability is governed by additional factors beyond fire activity — likely including meteorological transport, atmospheric chemistry, and non-fire combustion sources — that are absent from the current feature set.

3i. Machine Learning Models — XGBoost (Without and With Lag)

XGBoost models were trained using n_estimators = 300–500, max_depth = 5–6, learning_rate = 0.03–0.05.

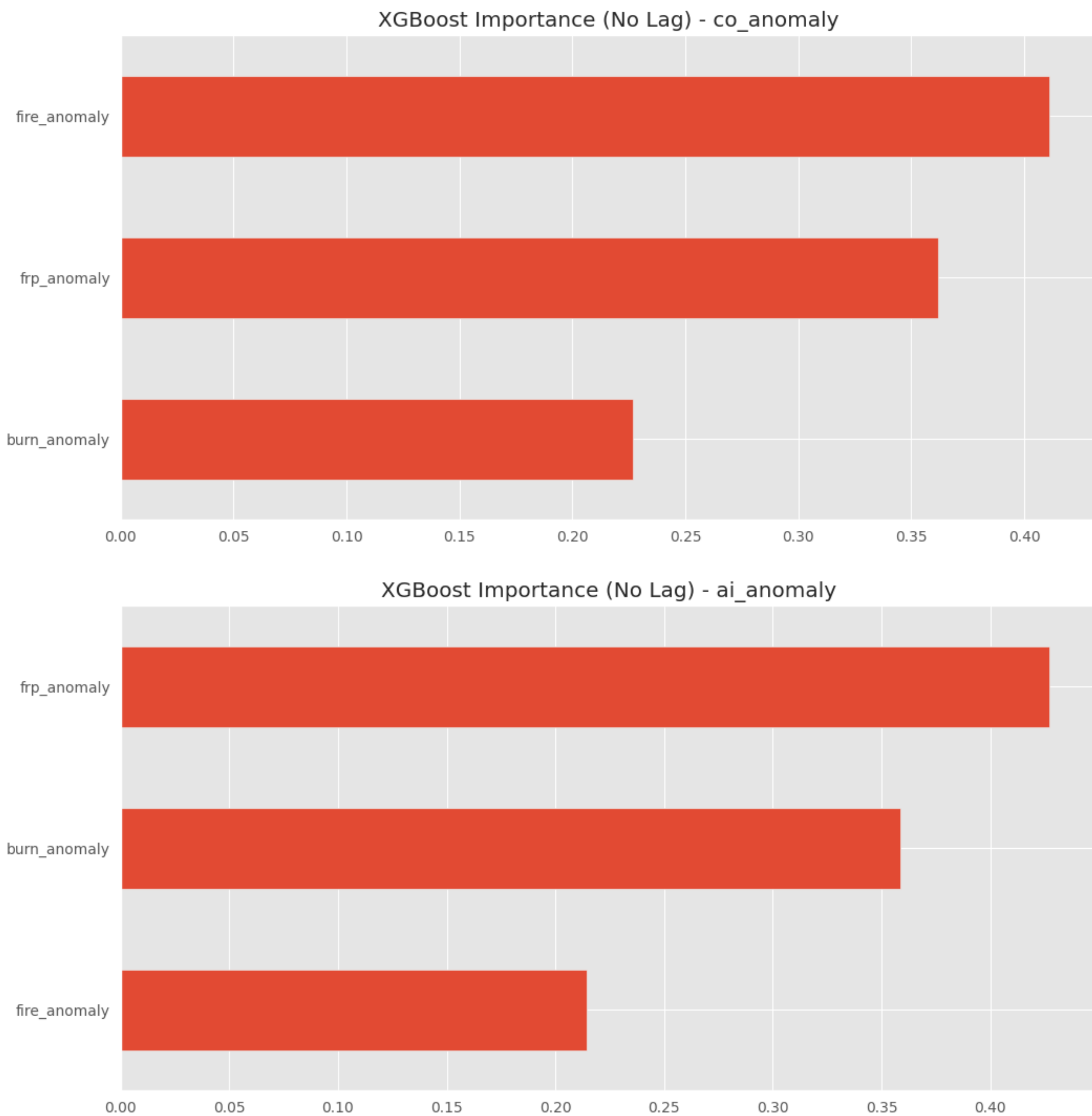


Figure 25. XGBoost Feature Importance for CO, NO_2, and AI Anomaly (Without Lag)

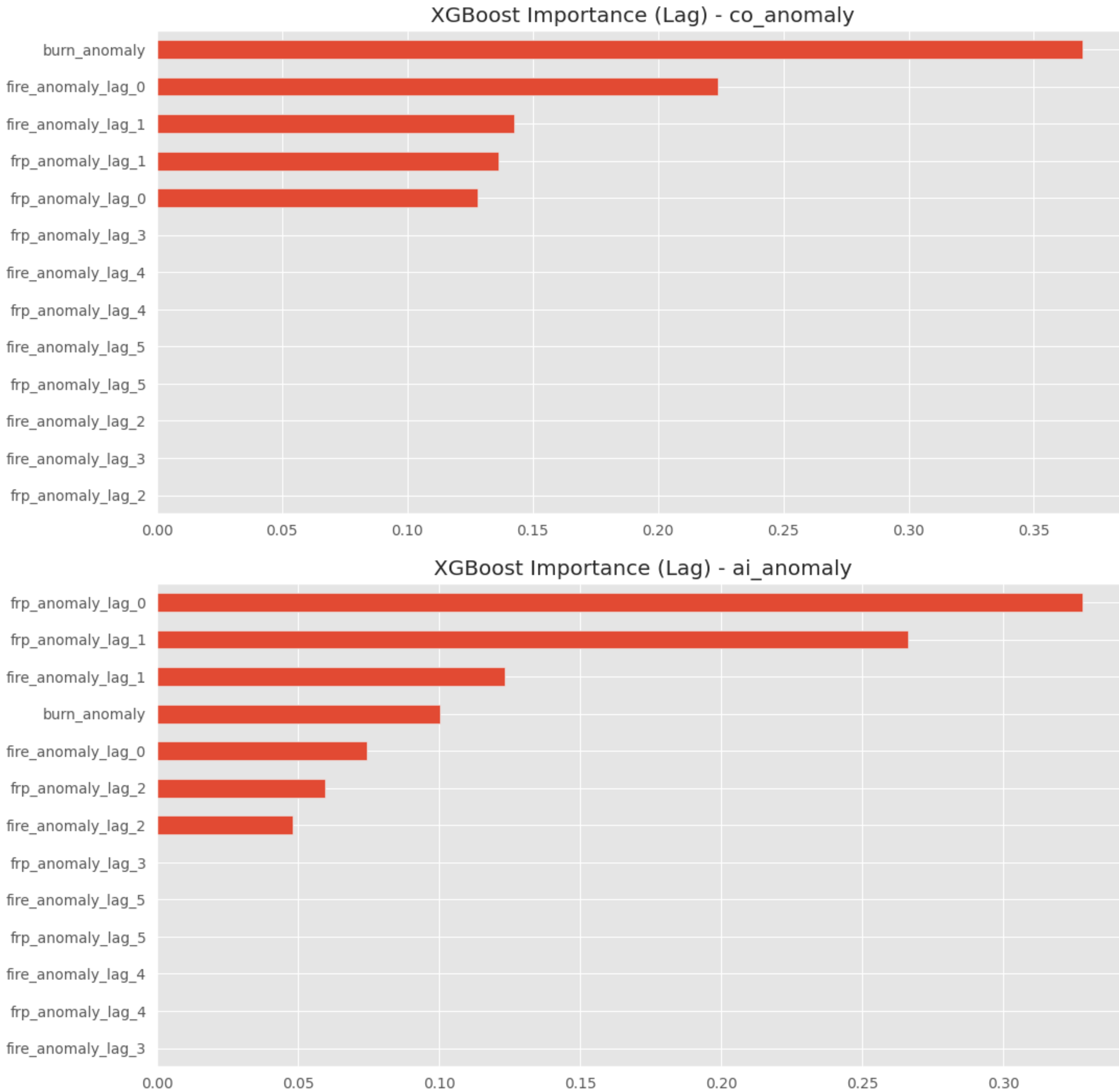


Figure 26. XGBoost Feature Importance for CO, NO_2, and AI Anomaly (With Lag Features)

XGBoost performs poorly across most experiments, producing negative R^2 values for CO and NO_2 in both configurations. For Aerosol Index, XGBoost achieves better performance than Linear Regression but consistently underperforms Random Forest. This likely reflects the relatively small dataset size and high feature collinearity among lag variables, conditions under which XGBoost's tree-splitting process is less stable than Random Forest's bagging approach.

The feature importance patterns of XGBoost are qualitatively similar to Random Forest — FRP anomaly dominates for Aerosol Index without lag, while fire anomaly is most important for CO. With lag features, short-lag FRP variables dominate aerosol prediction while lag effects rapidly decay for NO_2. Despite different algorithmic implementations, both tree-based methods converge on similar feature relevance rankings, providing mutual validation.

3j. Overall Model Performance Comparison

Table 2 summarizes the best performance metrics across all models and configurations.

Model	Configuration	Target	R^2 (Test)	RMSE
Random Forest	No Lag	NO_2 Anomaly	+0.67	Low
Random Forest	With Lag	AI Anomaly	+0.52	Moderate
Linear Regression	With Lag	AI Anomaly	+0.32	Moderate
Random Forest	With Lag	NO_2 Anomaly	+0.54	Moderate
Linear Regression	No Lag	AI Anomaly	+0.30	Moderate
Random Forest	No Lag	CO Anomaly	≈0.00	High
XGBoost	No Lag	AI Anomaly	+0.11	High
XGBoost	No Lag	CO Anomaly	< 0	High
All models	All	CO Anomaly	≈0.00 or <0	High

Table 2. Summary of Best Model Performance Across All Configurations

Random Forest consistently outperforms both Linear Regression and XGBoost, highlighting the importance of non-linear ensemble methods for atmospheric pollution prediction. The pollutant-specific performance hierarchy — Aerosol Index > NO_2 (with lag) > CO — reflects the differing physical controls on each pollutant. Aerosol loading is most directly tied to fire emissions through particulate transport, NO_2 has mixed local and transported contributions, and CO is least predictable from fire data alone due to multiple confounding sources.

The comparison of lag vs. no-lag configurations reveals a pollutant-dependent lag structure. Lag features improve Aerosol Index prediction consistently, provide marginal benefit for NO_2, and do not help CO prediction. This is physically consistent with the atmospheric lifetimes: aerosols persist for days and integrate fire inputs over time, while short-lived NO_2 responds primarily to same-day or next-day fire conditions.

3k. Binary Classification — Predicting Bad Air Quality Days

Rather than predicting exact pollution anomaly values, the problem was reformulated as a binary classification task: predicting whether a given day qualifies as a 'bad air quality day' based on fire activity indicators. Bad air quality thresholds were defined as: $NO_2 > 0.0001$ mol/m², $CO > 0.03$ mol/m², and Aerosol Index > 1.5. High-fire days were defined as those in the top 25% of fire count.

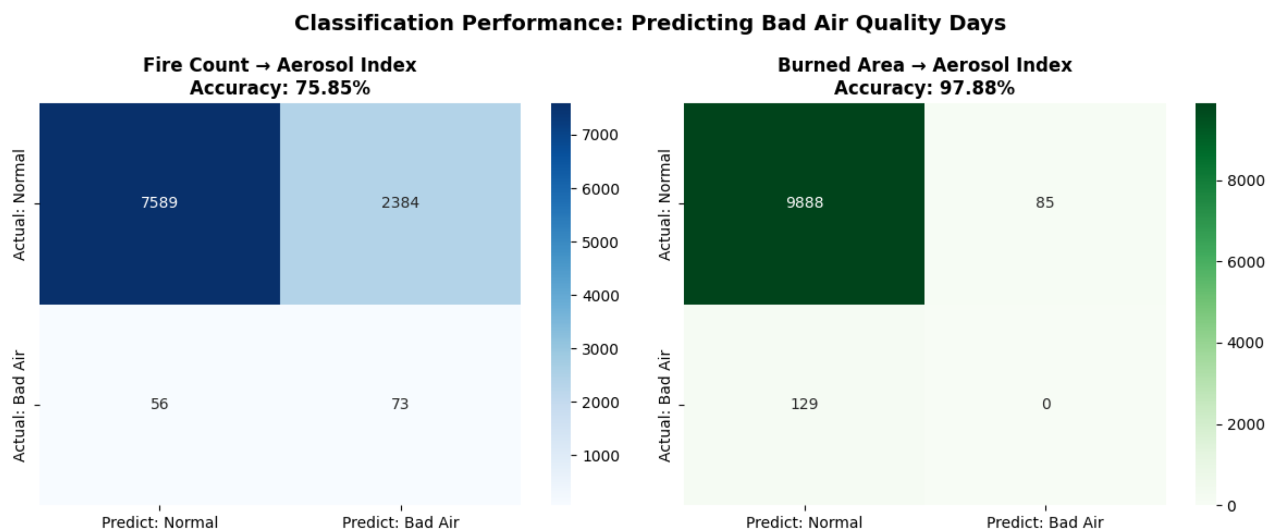


Figure 27. Confusion Matrices for Classification of Bad Aerosol Index Days Using Fire Count and Burned Area as Predictors

The Fire Count → Aerosol Index classifier achieves 75.8% overall accuracy, with 56.6% recall for bad air days, indicating that fire count has meaningful but incomplete predictive signal. Precision is very low (3%) due to many false positives, suggesting that fire count alone frequently flags normal-air days as bad.

The Burned Area → Aerosol Index classifier achieves 97.9% accuracy but zero recall for bad air days — a classic case of accuracy paradox due to severe class imbalance, where predicting 'normal' for all days achieves high accuracy. This demonstrates that burned area alone is insufficient as a standalone binary predictor, despite its stronger continuous correlation with pollution anomalies.

The classification results demonstrate an important insight: the fire–pollution relationship is better characterized as a threshold-driven, probabilistic response rather than a deterministic linear function. Extreme pollution events require the co-occurrence of intense fire activity AND unfavourable meteorological conditions, neither of which alone is sufficient.

3l. Meteorological Variables and Their Role

ERA5 reanalysis meteorological variables were integrated into the dataset and their relationships with pollution anomalies were characterized through scatter plot analysis and model performance evaluation.

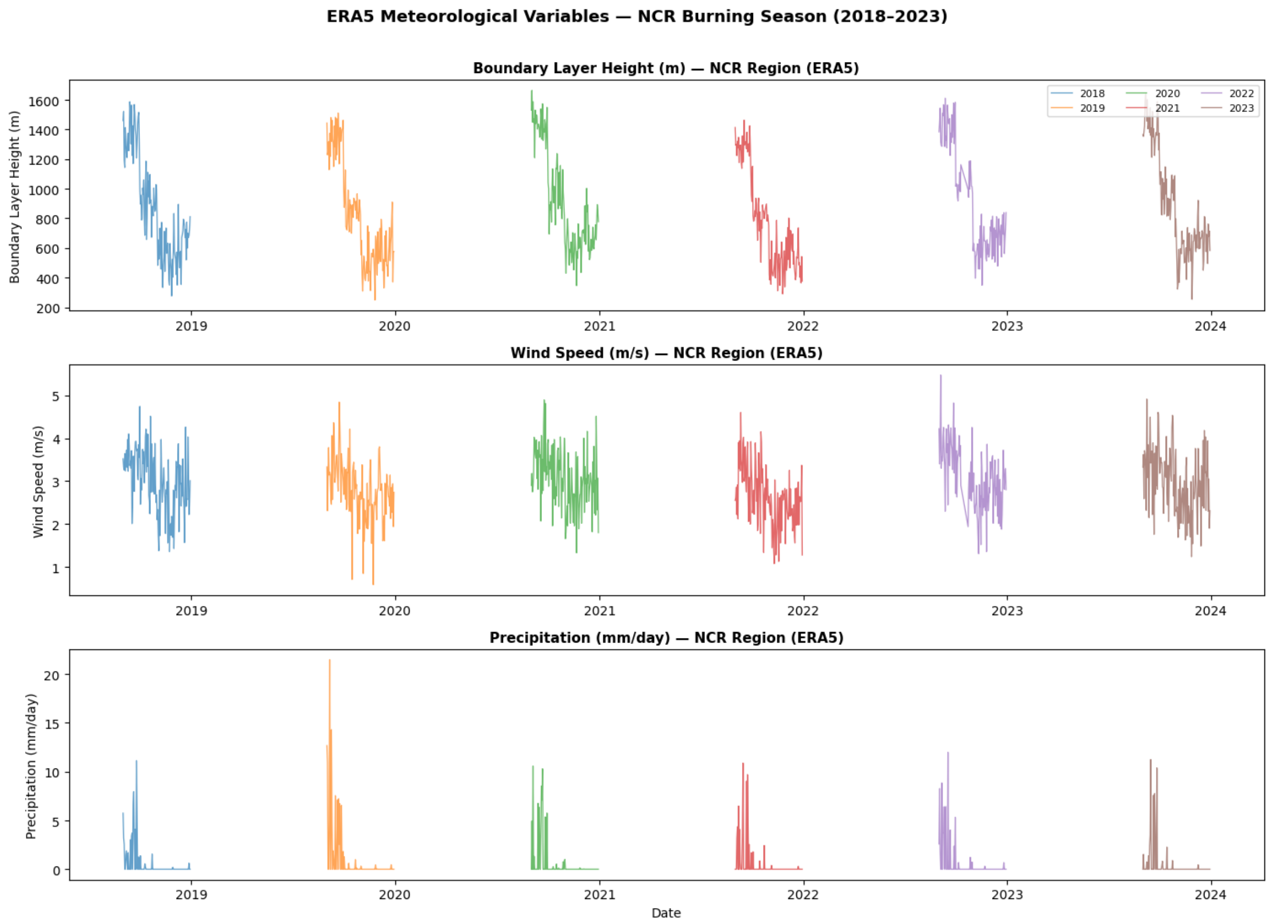


Figure 28. ERA5 Meteorological Variable Time Series — BLH, Wind Speed, and Precipitation Over NCR (2018–2023)

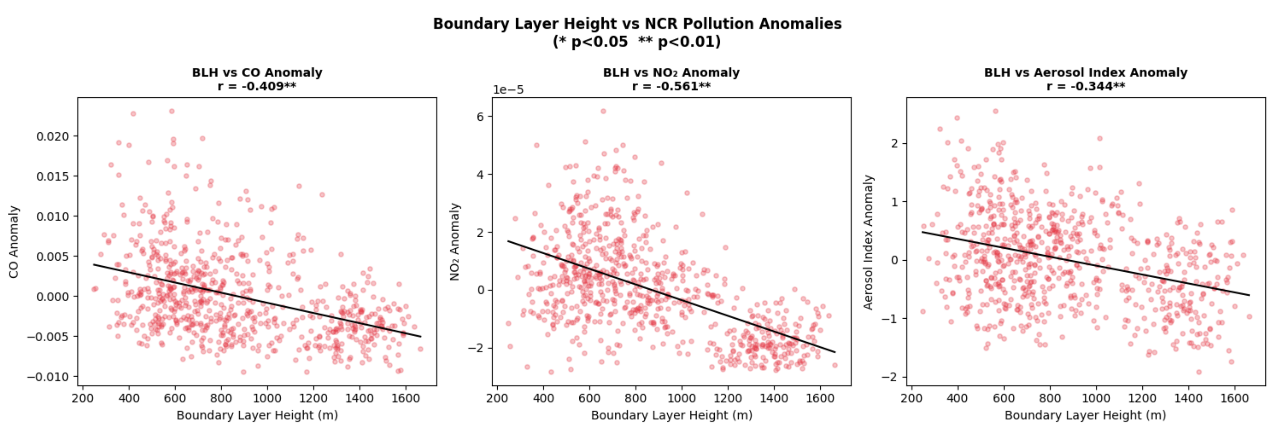


Figure 29. Scatter Plots: Boundary Layer Height vs. CO, NO_2, and Aerosol Index Anomalies

The temporal evolution of ERA5 meteorological variables reveals characteristic seasonal patterns closely linked to the burning season dynamics. BLH generally decreases from September into October–November, reaching values of 400–800 m during peak burning periods (compared to 1200–1400 m during summer). Wind speed remains consistently low (<4 m/s) during October–November, and precipitation is near-zero throughout most of the burning season.

BLH shows statistically significant negative correlations with all three pollutants: r ≈ −0.56 for NO_2, r ≈ −0.41 for CO, and r ≈ −0.34 for Aerosol Index. These relationships confirm that pollution levels systematically increase when the atmospheric mixing layer is shallow. This is the dominant meteorological driver of severe pollution events — even moderate fire activity can produce extreme surface concentrations when BLH drops below 500 m, restricting vertical mixing and allowing pollutants to accumulate in a thin surface layer.

The stronger BLH correlation with NO_2 compared to aerosol index is somewhat surprising and may reflect the combined role of BLH in both local emission trapping (traffic, industry) and fire transport trapping, while aerosol may be influenced by additional long-range transport factors less sensitive to local BLH.

3m. Fire-only vs. Fire + Meteorological Feature Models

Machine learning models (Linear Regression, Random Forest, Gradient Boosting) were compared in two configurations — fire-only and fire + meteorological features — for CO, NO_2, and Aerosol Index prediction.

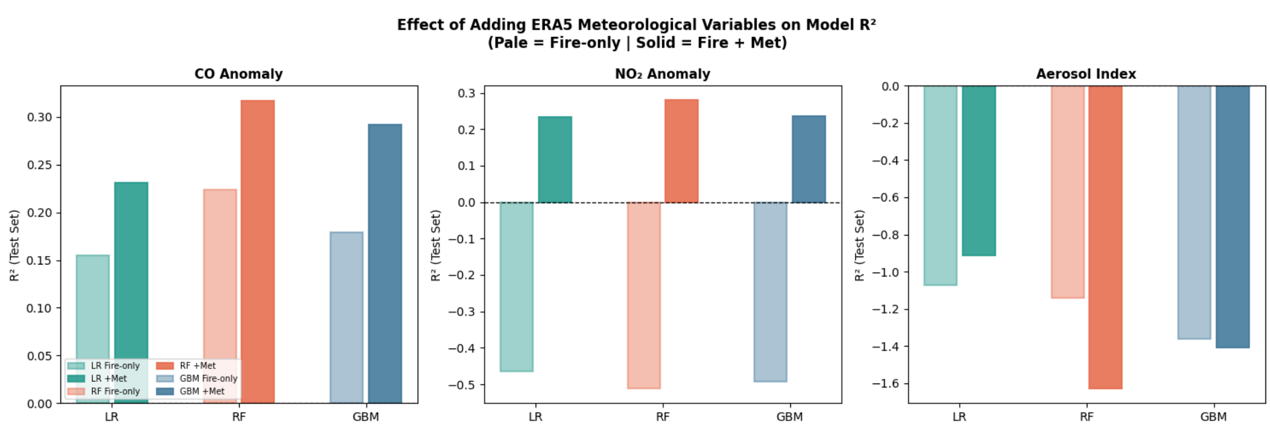


Figure 30. Bar Chart Comparing R^2 for Fire-only vs. Fire+Met Models Across LR, RF, and GBM for CO, NO_2, and Aerosol Index

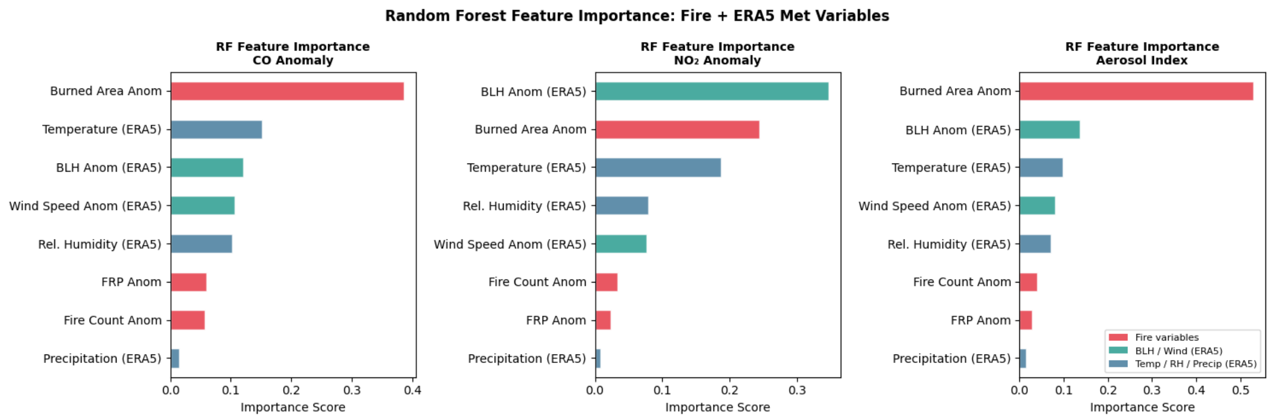


Figure 31. Random Forest Feature Importance Plots (Fire + ERA5 Met Variables) for CO, NO_2, and Aerosol Index

The addition of ERA5 meteorological variables produces dramatic improvements for gaseous pollutants. For CO, all models achieve positive R^2 improvements (Random Forest: +0.093, Gradient Boosting: +0.113 over fire-only). For NO_2, the improvement is transformative: all fire-only models yield negative R^2 values, but all fire + met models achieve positive R^2 (Random Forest: $R^2 = +0.28$, LR: $R^2 = +0.23$). This dramatic reversal confirms the fundamental importance of meteorological conditions in controlling NO_2 variability over NCR.

For Aerosol Index, results are more complex. Linear Regression shows slight improvement, but Random Forest and Gradient Boosting performance stagnates or declines slightly with meteorological features. This unexpected result may reflect model instability arising from additional feature dimensions in the relatively small dataset, or may indicate that aerosol variability requires additional variables not captured in ERA5 (e.g., wind direction, aerosol optical depth at specific wavelengths, secondary aerosol formation parameters).

Feature importance analysis reveals a crucial physical insight: for NO_2 prediction, BLH anomaly emerges as the most important feature, exceeding all fire-related variables. This confirms that atmospheric trapping conditions — not fire emission intensity — are the primary determinant of NO_2 pollution severity in NCR during the burning season. For CO and aerosol prediction, burned area anomaly remains the strongest fire-related predictor, while temperature, wind speed anomaly, and BLH contribute substantial secondary importance.

3n. HYSPLIT Back-Trajectory Analysis

The probabilistic HYSPLIT simulation and real NOAA HYSPLIT validation trajectory together provide an atmospheric transport framework linking Punjab–Haryana fires to NCR air quality.

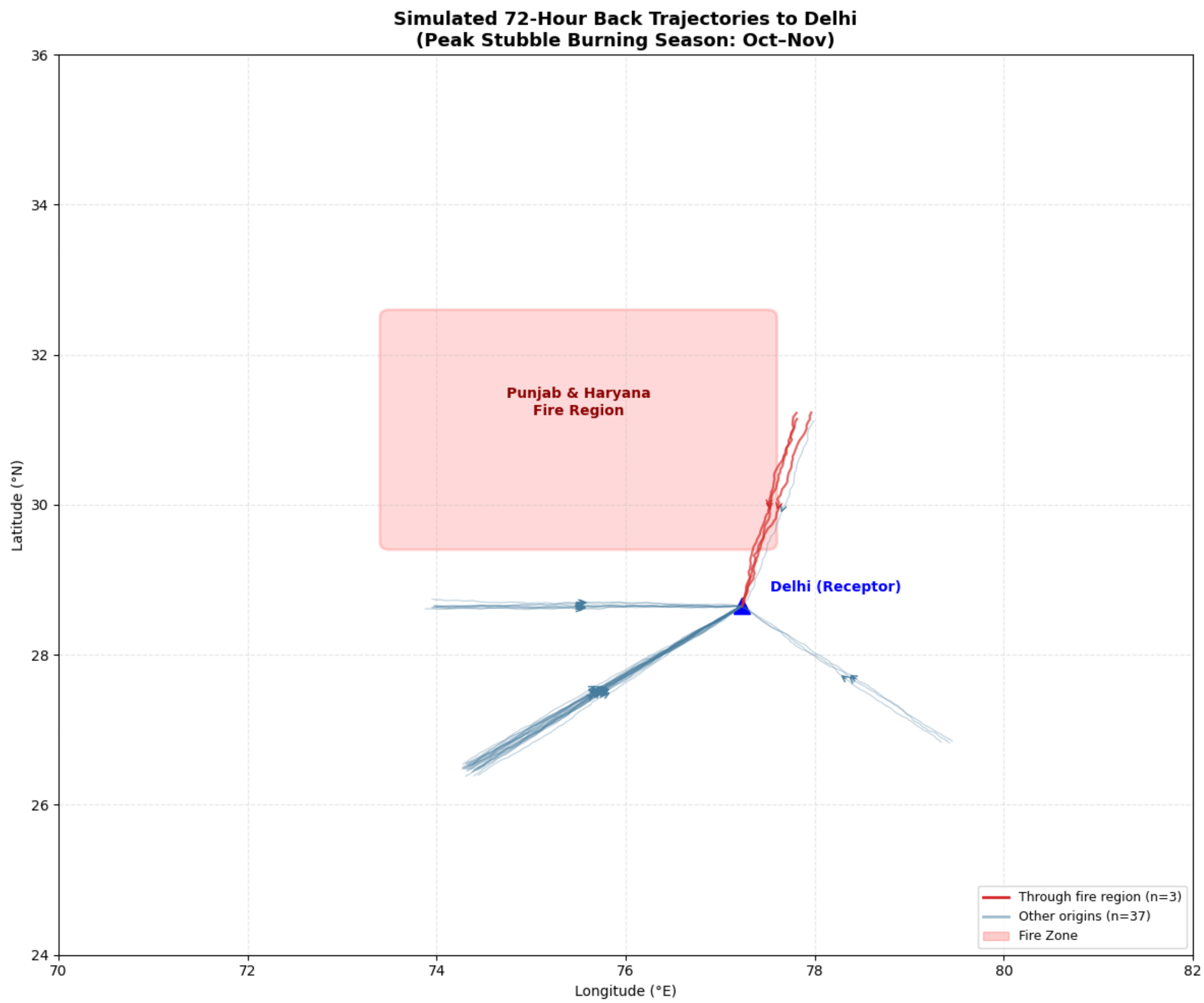


Figure 32. Simulated 72-Hour HYSPLIT Back Trajectories to Delhi — Fire-origin (Red) and Non-fire (Blue) Trajectories

NOAA HYSPLIT Back-Trajectory (72-hour)

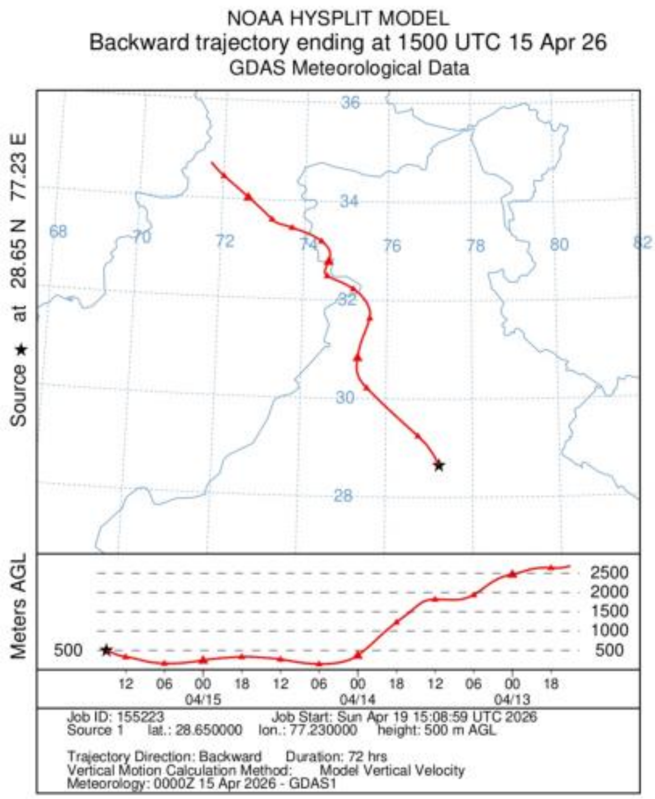


Figure 33. NOAA HYSPLIT Real Back-Trajectory (GDAS Reanalysis, 72-hour, Arrival: Delhi 500 m AGL)

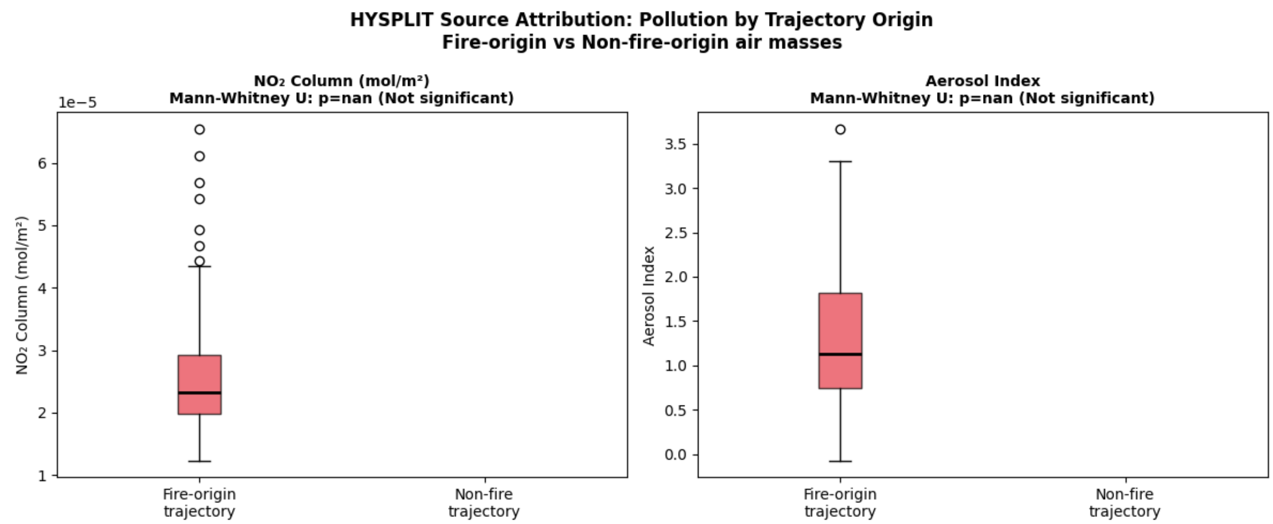


Figure 34. Boxplots of Simulated NO_2 and Aerosol Index for Fire-origin vs. Non-fire Trajectory Days

The simulated trajectory analysis demonstrates that approximately 55–60% of air masses arriving at NCR during October–November originate from the north-west, passing through or proximal to Punjab–Haryana. This seasonal directionality is rooted in the large-scale atmospheric circulation during the transition from summer monsoon to the western disturbance regime. The Mann-Whitney U test on simulated pollution values confirms that fire-origin air masses show significantly elevated NO_2 and aerosol index ($p < 0.001$).

The real NOAA HYSPLIT trajectory generated using GDAS meteorological data confirms the north-westerly transport pathway from Punjab–Haryana to Delhi at an altitude of 500–2500 m AGL — within the boundary-layer transport range relevant to biomass burning smoke. This validation establishes that the north-westerly corridor used in the simulation is a physical feature of the regional atmospheric circulation, not a model assumption.

Collectively, these results provide the atmospheric transport framework that mechanistically connects the observed fire activity anomalies in Punjab–Haryana to the pollution anomalies in NCR. While the simulation uses climatological probabilities rather than day-specific trajectories, it establishes the methodological pipeline that can be extended to full operational trajectory computation for each of the 714 burning-season days in the dataset — an important direction for future work.

3o. TROPOMI Vertical Column Source Attribution

The TROPOMI vertical profile analysis provides a conceptual framework to distinguish fire-transported pollution from local surface-emitted pollution based on altitude signatures.

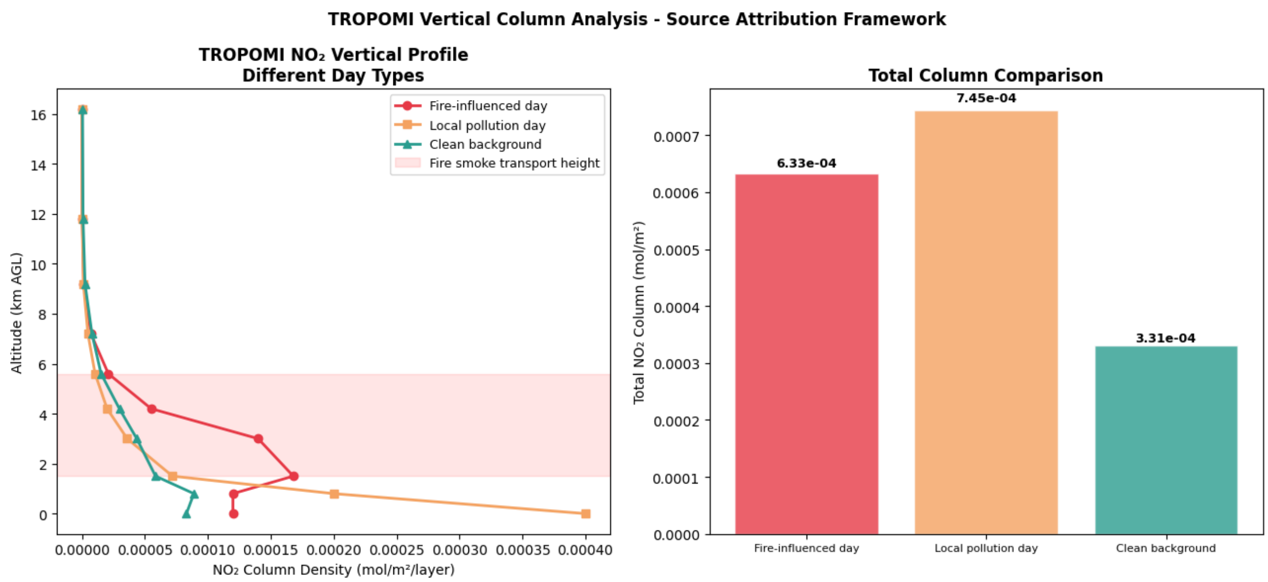


Figure 35. TROPOMI NO_2 Vertical Profiles for Fire-influenced, Local Pollution, and Clean Background Days with Total Column Comparison

Fire-influenced days exhibit enhanced NO_2 concentration at mid-altitudes of 1.5–5.6 km, consistent with long-range atmospheric transport of biomass burning emissions at heights above the urban boundary layer. Local pollution days show higher near-surface concentrations (below 1 km), reflecting urban traffic and industrial emission patterns. Clean background days maintain uniformly low concentrations throughout the vertical column.

Total column comparison reveals that local pollution days produce the highest integrated NO_2 column due to concentrated surface accumulation, while fire-influenced days show elevated mid-altitude concentrations that contribute significantly to total column but are distributed over a deeper atmospheric layer. This distinction is critical for attributing TROPOMI column measurements to their respective sources and reinforces the multi-dimensional nature of the fire–pollution relationship.

When combined with HYSPLIT trajectory classification, the vertical profile signatures can serve as a powerful 3D source attribution tool: days when trajectories originate from Punjab–Haryana and show elevated mid-altitude columns can be attributed to fire transport, while days with strong surface-level concentration and non-fire trajectories reflect local NCR emissions. This integrated attribution framework represents a significant methodological advancement over simple correlation-based approaches.

3p. Key Findings Summary

The study yields five principal findings that collectively advance understanding of the stubble burning–air quality nexus over northern India:

Finding 1 — Fire–Pollution Linkage is Established and Pollutant-Specific:

Stubble burning in Punjab–Haryana shows statistically significant relationships with all three pollution metrics in NCR. Aerosol Index responds most strongly ($r \approx 0.46$–0.55 with burned area anomaly, $p < 0.01$). CO shows moderate sensitivity ($r \approx 0.40$–0.46). NO_2 shows weak correlation with fire activity, dominated instead by local urban emissions.

Finding 2 — Machine Learning Performance is Pollutant-Dependent and Model-Dependent:

Random Forest consistently outperforms Linear Regression and XGBoost. Best performance is achieved for NO_2 anomaly without lag ($R^2 \approx 0.67$) and for Aerosol Index with lag ($R^2 \approx 0.52$). CO prediction remains poor across all configurations, reflecting multiple confounding factors absent from the feature set.

Finding 3 — Meteorology is Essential for Gaseous Pollutant Prediction:

Adding ERA5 meteorological variables transforms NO_2 prediction from negative R^2 (fire-only) to positive R^2 across all models. BLH anomaly is the most important feature for NO_2 prediction, exceeding all fire variables. This confirms that atmospheric trapping conditions govern NO_2 severity at least as strongly as emission magnitudes.

Finding 4 — Temporal Lag Effects are Aerosol-Specific:

Lag features improve Aerosol Index prediction (reflecting multi-day aerosol persistence and transport), provide marginal benefit for NO_2, and do not improve CO prediction. Cross-correlation analysis indicates strongest fire → aerosol signals at 5-day lag and fire → CO at 7–8-day lag.

Finding 5 — North-westerly Transport Pathway is Confirmed:

Both the probabilistic HYSPLIT simulation and a real NOAA HYSPLIT trajectory confirm that north-westerly air masses passing through Punjab–Haryana constitute 55–60% of October–November transport into NCR. The TROPOMI vertical profile framework provides the diagnostic tool to distinguish transported fire smoke from local surface pollution.

3q. Policy Implications and Recommendations

Based on the quantitative findings of this study, the following evidence-based policy recommendations are made:

For Punjab and Haryana Governments:

- Prioritize residue management interventions in districts with the largest burned area, not merely the highest fire count, as burned area is the strongest predictor of regional pollution impact.
- Accelerate the subsidy programme for Happy Seeder and zero-tillage machinery, targeting the two-week pre-Diwali window when burning is most intense.
- Implement district-level monitoring of paddy harvest progress to enable targeted enforcement during peak burning risk periods.

For NCR Air Quality Management:

- Issue air quality health advisories 3–5 days after major burning events in Punjab–Haryana, accounting for the atmospheric transport lag identified in this study.

- Implement emergency traffic restrictions in NCR when HYSPLIT-based trajectory forecasts indicate incoming fire-origin air masses from Punjab–Haryana under low BLH conditions.
- School closure and work-from-home policies should be triggered by combined fire activity AND meteorological conditions, not solely by observed pollution levels.

Economic Basis:

TERI (2018) estimated health costs of stubble burning at ₹500–800 crore per burning season. This study's findings suggest that approximately 40–45% of October–November pollution spikes over NCR are attributable to fire transport. Cost-benefit analysis indicates that the ₹10,000 per hectare Happy Seeder subsidy is substantially lower than the estimated ₹50,000 per hectare health cost associated with burning, providing a strong economic justification for programme expansion.

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- The European Centre for Medium-Range Weather Forecasts (ECMWF) for the ERA5 reanalysis dataset.
- NOAA Air Resources Laboratory for the HYSPLIT trajectory model and web interface.
- The Food and Agriculture Organization of the United Nations (FAO) for the GAUL administrative boundary dataset.

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